

Erdős Problem 848: Exact Hall Closure by Square-Sieve Descent and Prefix Colouring

Alex Chengyu Li

Independent Researcher, Tokyo, Japan

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Abstract

For $N \geq 1$, let $A \subseteq [1, N]$ have the property that $ab + 1$ is nonsquarefree for every $a, b \in A$. We prove that

$$|A| \leq |\{n \leq N : n \equiv 7 \pmod{25}\}|,$$

with equality attained by the progression $7 \pmod{25}$. The proof first rewrites the assertion as an exact Hall inequality. An exact prefix colouring proves the inequality through $1.5 \cdot 10^6$; two finite square-sieve arguments cover the interval up to $5 \cdot 10^6$; and a valuation-cell-fibre descent covers the four intervals $[5, 10)$, $[10, 20)$, $[20, 40)$, and $[40, \infty)$ million. All finite enumerations used in the proof employ integer or rational arithmetic and are reproduced by the accompanying source files; their semantic checkers and the terminal all- N theorem are replayed by the ordinary Lean kernel.

1 Statement and Hall formulation

Write

$$C_N = \{n \in [1, N] : n \equiv 7 \pmod{25}\}.$$

We prove

$$|A| \leq |C_N| \tag{1}$$

for every N and every $A \subseteq [1, N]$ such that $ab + 1$ is nonsquarefree for all $a, b \in A$. Sawhney proved the assertion for all sufficiently large N by a density sieve and structural reduction [1]; the argument below supplies an explicit all- N proof.

Proposition 1.1 (Sharpness). *The set C_N is admissible: if $a, b \in C_N$, then $ab + 1$ is not squarefree.*

Proof. Since $a \equiv b \equiv 7 \pmod{25}$,

$$ab + 1 \equiv 7^2 + 1 = 50 \equiv 0 \pmod{25}.$$

Thus $5^2 \mid ab + 1$. □

2 The direct Hall cut

Join $b \in [1, N] \setminus C_N$ to $a \in C_N$ when $ab + 1$ is squarefree. For $B \subseteq [1, N] \setminus C_N$, write

$$\Gamma_N(B) = \{a \in C_N : ab + 1 \text{ is squarefree for some } b \in B\}$$

and

$$T_N(B) = C_N \setminus \Gamma_N(B).$$

Thus $a \in T_N(B)$ precisely when $ab + 1$ is non-squarefree for every $b \in B$.

For every set $B \subseteq [1, N] \setminus C_N$ such that $bc + 1$ is nonsquarefree for all $b, c \in B$, the Hall inequality to be proved is

$$|B| + |T_N(B)| \leq |C_N|, \quad (2)$$

equivalently $|\Gamma_N(B)| \geq |B|$.

Theorem 2.1 (Hall reduction). *If the Hall inequality (2) holds, then the exact bound (1) holds for every N .*

Proof. Let A be admissible and put $B = A \setminus C_N$. Every subset $D \subseteq B$ is again a compatible outside set, so $|\Gamma_N(D)| \geq |D|$ by the Hall inequality (2). Hall's marriage theorem gives an injection from B into $\Gamma_N(B)$. Admissibility gives

$$\Gamma_N(B) \cap (A \cap C_N) = \emptyset,$$

because every vertex of $\Gamma_N(B)$ has a squarefree edge to some element of $B \subseteq A$. Hence

$$|B| \leq |\Gamma_N(B)| \leq |C_N \setminus A|,$$

and therefore

$$|A| = |A \cap C_N| + |B| \leq |A \cap C_N| + |C_N \setminus A| = |C_N|.$$

□

Theorem 2.2 (Hall equivalence). *the Hall inequality (2) is equivalent to the exact Erdős 848 upper bound (1).*

Proof. The forward implication is Theorem 2.1. Conversely, assume (1) and let B satisfy the hypotheses of the Hall inequality (2). Put

$$A = B \cup T_N(B).$$

Products internal to B are non-squarefree by hypothesis. Products internal to $T_N(B)$ are divisible by 25, because all its elements are $7 \pmod{25}$. Finally, every cross-product between B and $T_N(B)$ is non-squarefree by the definition of $T_N(B)$. Hence A is admissible. Moreover B and $T_N(B)$ are disjoint, so (1) gives

$$|B| + |T_N(B)| = |A| \leq |C_N|,$$

which is (2). □

Thus the Hall inequality is the original theorem written at the exact exchange boundary. All subsequent estimates prove (2).

3 An exact finite prefix close

The whole finite remainder admits a single exact colouring certificate.

Theorem 3.1 (Two-matching prefix colouring). *For every $1 \leq N \leq 1.5 \cdot 10^6$, let*

$$\mathcal{D}_N = \{x \leq N : x^2 + 1 \text{ is nonsquarefree}\},$$

and join two distinct members $x, y \in \mathcal{D}_N$ when $xy + 1$ is nonsquarefree. This graph has a proper colouring with exactly $|\mathcal{C}_N|$ colours. Consequently (1) and the Hall inequality (2) hold throughout this interval.

Proof. Every member of an admissible set belongs to $\mathcal{D}_N$, and every two distinct members are joined. Thus it is enough to construct the asserted proper colouring.

Put $L = 1.5 \cdot 10^6$. The diagonal set $\mathcal{D}_L$ is enumerated without factoring L separate integers. If $p^2 \mid x^2 + 1$, then p is odd, $p \equiv 1 \pmod{4}$, and $p \leq x \leq L$. Conversely every solution of $z^2 \equiv -1 \pmod{p^2}$ marks an arithmetic progression of diagonal vertices. For each prime $p \leq L$, sieve primality, find the two roots modulo p , lift them uniquely modulo p^2 , verify the lifted congruence, and mark both progressions. This is an exact equivalence.

Process the marked vertices in increasing order. At a given prefix write

$$U = \{u : u \equiv 7 \pmod{25}\}, \quad V = \{v : v \equiv 18 \pmod{25}\}, \quad E = \mathcal{D}_N \setminus (U \cup V).$$

Maintain two injections into the colour set U . The first is $\mu : V \rightarrow U$ with

$$v\mu(v) + 1 \text{ squarefree.}$$

The second is $\nu : E \rightarrow U$, and it satisfies

$$e\nu(e) + 1 \text{ squarefree}, \quad e\mu^{-1}(\nu(e)) + 1 \text{ squarefree}$$

whenever the second factor exists. Then for every $u \in U$ the class

$$\{u\} \cup \mu^{-1}(u) \cup \nu^{-1}(u)$$

has at most three vertices, and every product plus one inside it is squarefree. These are therefore $|U| = |\mathcal{C}_N|$ independent colour classes.

The two injections are maintained prefix by prefix, using no future vertex. When a new 7-class vertex appears, add its singleton colour. When a new 18-class vertex appears, either put it in the unique colour not yet used by μ , or exchange it with one resident 18-class vertex; both new products in the exchange are checked. On the complete interval there are 39290 direct insertions and 20710 one-exchange insertions, and no longer augmenting path is used. An exceptional vertex is placed in a colour not yet used by ν after checking its one or two required products. If an exchange in μ invalidates a resident exceptional vertex, that vertex is immediately reinserted. There are 40692 such insertions or repairs: 40688 use a free colour directly, while only the initial vertices

$$41, \quad 70, \quad 99, \quad 117$$

use augmenting paths, of maximum depth four. Hence the two injections, and therefore the proper colouring, exist after every marked vertex is processed. Between marked vertices the

graph is unchanged, while a new colour appears exactly at a new member of C_N . The certificate consequently covers every prefix, not merely the upper endpoint.

The terminal exact counts are

quantity	value	quantity	value
primes through L	114155	lifted roots inspected	114044
$ \mathcal{D}_L $	157733	$ U = V $	60000
$ E $	37733	pure edge tests	125789
exceptional edge tests	88407	final verification pairs	135466.

All 270784 construction-oracle product queries are factored exactly. The terminal verifier then passes all 135466 class pairs through a fresh factorization oracle; in this run their products are distinct. Deterministic 64-bit Miller–Rabin verifies every factor as prime, their product is reconstructed, and repeated factors are rejected; 128-bit modular multiplication prevents overflow. A final pass checks one diagonal witness for every marked vertex, checks that no unmarked vertex was coloured, and refactors every pair within every terminal colour class. Compilation fails if assertions are disabled. The complete producer and verifier are `scripts/systematic_finite_prefix_coloring.cpp`.

Thus every admissible set is a clique in a graph properly coloured with $|C_N|$ colours, and so has at most $|C_N|$ members. Hall equivalence, Theorem 2.2, gives the last assertion. $\square$

4 An aggregate explicit large- N close

We now make the known sufficiently-large- N part numerically explicit and sharpen its threshold. We retain Sothanaphan’s explicit progression estimate [2], but aggregate the 23 non-base residue classes before estimating the diagonal Pell tail. This makes the large-square tail a single set of integers rather than 23 separately charged copies.

Let

$$C_{2,5} = 1 - \frac{25}{3\pi^2} < 0.155656803, \quad C_5 = 1 - \frac{25}{4\pi^2} < 0.366742603,$$

and

$$C_{\text{quad}} = 1 - \prod_{\substack{p \equiv 1 \pmod{4} \\ p \geq 13}} \left(1 - \frac{2}{p^2}\right) < 0.027346508.$$

Exact rational multiplication through 10^7 gives

$$1 - \prod_{\substack{13 \leq p \leq 10^7 \\ p \equiv 1 \pmod{4}}} \left(1 - \frac{2}{p^2}\right) < 0.027346490.$$

For the omitted factors,

$$1 - \prod_{p > 10^7} \left(1 - \frac{2}{p^2}\right) \leq 2 \sum_{p > 10^7} \frac{1}{p^2} < 0.000000017856.$$

For $X = 10^7$, partial summation, $\pi(X) = 664579$, and $\pi(t) < 1.25506t/\log t$ give

$$2 \sum_{p > X} \frac{1}{p^2} \leq -\frac{2\pi(X)}{X^2} + 4(1.25506) \int_X^\infty \frac{dt}{t^2 \log t} < 0.000000017856.$$

Thus the displayed inequality retains the negative boundary term, and the two strict bounds prove the asserted value of C_{quad} . Fix $q_0 \in \{25, 100\}$ and $b \leq N$. For every residue class t such that $p^2 \nmid bt + 1$ for every prime $p \mid q_0$, the explicit off-diagonal progression estimate gives

$$\#\{a \leq N : a \equiv t \pmod{q_0}, \mu(ab + 1) = 0\} \leq \frac{N}{q_0} \left(1 - \prod_{(p, q_0 b) = 1} \left(1 - \frac{1}{p^2} \right) \right) + \Phi_{q_0} N^{3/4}, \quad (3)$$

where $\Phi_{25} = 4.7$ and $\Phi_{100} = 5.3$. For completeness, the source's normalized error function is

$$\begin{aligned} E_q(N) = & \frac{2.367}{N^{3/4}} + \alpha_q + \frac{1}{N^{3/4}} + \frac{N/q + 1}{\alpha_q N^{3/2}} + \frac{1}{\alpha_q \sqrt{q} N^{3/2}} \\ & + \frac{2\lambda q^{1/4}(N/q + 1)}{\alpha_q N^{5/4}} + \frac{2\lambda}{\alpha_q q^{3/8} N^{3/2}} + \frac{\lambda q^{5/4}(N/q + 1)}{\alpha_q^2 N} + \frac{\lambda q^{1/4}}{\alpha_q^2 N^2}, \end{aligned}$$

where

$$\alpha_{25} = \frac{25}{8}, \quad \alpha_{100} = \frac{7}{2}, \quad \lambda = \frac{128}{(8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} < 6.876.$$

It bounds the coefficient of $N^{3/4}$ in (3). Every nonconstant term decreases with N , and direct substitution at $N = 1.6 \cdot 10^{15}$ gives

$$E_{25}(1.6 \cdot 10^{15}) < 4.69949, \quad E_{100}(1.6 \cdot 10^{15}) < 5.27503.$$

Thus (3) with the displayed rounded constants is valid throughout the range used below, not only at the older published threshold.

The root-count factor in this estimate can be made much smaller in the present application. Let $\rho(m)$ be the maximum number of roots of $y^2 \equiv u \pmod{m}$ over units u , and let ℓ_j be the j -th odd prime, $P_k = \prod_{j \leq k} \ell_j$. Put

$$\begin{aligned} Q_* &= 8P_{12} = 1217001054108840, \\ N_{100} &= Q_*/20 = 60850052705442, \quad N_{25} = Q_*/5 = 243400210821768. \end{aligned}$$

For $7.67769 \cdot 10^9 \leq N \leq 1.6 \cdot 10^{15}$ define

$$H_{25}(N) = \begin{cases} 8192, & N < N_{25}, \\ 16384, & N \geq N_{25}, \end{cases} \quad H_{100}(N) = \begin{cases} 8192, & N < N_{100}, \\ 16384, & N \geq N_{100}. \end{cases}$$

For $q_0 \in \{25, 100\}$, $H = H_{q_0}(N)$, set

$$d_H = (2H)^{1/3} N^{2/3}$$

and define the normalized error envelope

$$\begin{aligned} \mathcal{E}_{q_0, H}(N) = & \frac{1}{N} \left(2.367 + d_H + 1 + \frac{(1 + 2H)(N/q_0 + 1)}{d_H} + \frac{1 + 2H}{\sqrt{q_0} d_H} \right. \\ & \left. + \frac{H q_0 N (N/q_0 + 1)}{d_H^2} + \frac{H}{d_H^2} \right). \end{aligned} \quad (4)$$

Lemma 4.1 (Primorial progression error). *In (3), throughout $7.67769 \cdot 10^9 \leq N \leq 1.6 \cdot 10^{15}$, the term $\Phi_{q_0} N^{3/4}$ may be replaced by*

$$N\mathcal{E}_{q_0, H_{q_0}(N)}(N).$$

On every interval on which the corresponding H_{q_0} is constant, the error envelope is decreasing.

Proof. In the source reduction the relevant modulus is $d_1 = q_0 b/g$, where g divides the radical of q_0 . Removing one redundant factor 5 when its exponent is at least two shows that there is an $m \leq q_0 N/5$ with $\rho(m) = \rho(d_1)$. If $m = 2^\nu m_0$ with m_0 odd, then

$$\rho(m) \leq c_\nu 2^{\omega(m_0)}, \quad c_0 = c_1 = 1, \quad c_2 = 2, \quad c_\nu = 4 \quad (\nu \geq 3).$$

The least modulus with root count greater than 8192 is $8P_{12} = Q_*$. The least one with root count greater than 16384 is $8P_{13} = 52331045326680120$, which exceeds $20(1.6 \cdot 10^{15})$. This proves the two step-function bounds for $\rho(d_1)$.

Repeat the Möbius split underlying (3), but keep $\rho(d_1) \leq H$ instead of replacing it by $6.876d_1^{1/4}$. With $X \leq N/q_0 + 1$, $\sqrt{q_0} \leq d_1 \leq q_0 N$, and $D = \lceil d_H \rceil$, the total additive error is at most

$$2.367 + D + \frac{(1 + 2H)X}{D} + \frac{1 + 2H}{\sqrt{q_0}D} + \frac{Hq_0NX}{D^2} + \frac{H}{D^2}.$$

Use $d_H \leq D \leq d_H + 1$, increasing the positive D term and decreasing all denominators, to obtain (4). Every normalized term is decreasing while H is fixed. $\square$

For the diagonal part put

$$A_7 = \{a \in A : a \equiv 7 \pmod{25}\}, \quad A_{18} = \{a \in A : a \equiv 18 \pmod{25}\}, \quad A^* = A \setminus (A_7 \cup A_{18}),$$

and define

$$U_{25} = \{x \leq N : x \not\equiv 7, 18 \pmod{25}\}, \quad U_{50} = \{x \in U_{25} : x \text{ is odd}\}.$$

For $r \in \{0, 1, 2, 3\}$ also put

$$U_{100,r} = \{x \in U_{25} : x \equiv r \pmod{4}\};$$

this is a union of 23 classes modulo 100. Two coprimality moduli will be used. Put

i	M_i	$\eta_i = \varphi(M_i)/M_i$	$h_i > \sum_{\substack{1 \leq a \leq M_i \\ (a, M_i)=1}} 1/a$
0	2310	16/77	2.189
1	43890	288/1463	2.685

and $e_0 = 64$, $e_1 = 128$. For $R \geq M_i^2$, put

$$K_i(R) = 2\eta_i R \left(h_i + \eta_i \left(1 + \log \frac{\sqrt{R}}{M_i} \right) \right) + e_i \sqrt{R}.$$

For $q \in \{25, 50, 100\}$ write $\theta_q = 23/q$, put

$$r_N = \frac{12}{5} N^{2/3},$$

and set, whenever $r_N \geq M_i^2$,

$$\begin{aligned} \Delta_q^{(i)}(N) &= \frac{23}{N} + \frac{3K_i(r_N + 1)}{N} + \frac{46(\log(r_N + 1) + 2)}{qr_N} \\ &\quad + \frac{N^2 + 1}{Nr_N^2} \left(1 + \log_{2+\sqrt{3}} N\right). \end{aligned} \quad (5)$$

Lemma 4.2 (Aggregate diagonal estimate). *For $q = 25$ and $U_q = U_{25}$, for $q = 50$ and $U_q = U_{50}$, or for $q = 100$ and $U_q = U_{100,r}$ with any fixed $r \pmod{4}$,*

$$\#\{x \in U_q : \mu(x^2 + 1) = 0\} \leq N(\theta_q C_{\text{quad}} + \Delta_q^{(i)}(N)) \quad (i = 0, 1), \quad (6)$$

provided $r_N \geq M_i^2$. The $i = 0$ envelopes decrease for $N \geq 7.8 \cdot 10^{10}$, and at that endpoint

$$\Delta_{25}^{(0)}(N) < 0.002663263, \quad \Delta_{50}^{(0)}(N) < 0.002662851, \quad \Delta_{100}^{(0)}(N) < 0.002662646.$$

The $i = 1$ envelopes decrease for $N \geq 3 \cdot 10^{13}$, and at that endpoint

$$\Delta_{25}^{(1)}(N) < 0.000402450, \quad \Delta_{50}^{(1)}(N) < 0.000402441.$$

Proof. Let $R = \lceil r_N \rceil$. In the Möbius expansion of the indicator of $x^2 + 1$ being squarefree, only squarefree s composed of primes $p \equiv 1 \pmod{4}$, $p \geq 13$, contribute. For such an s , the congruence $x^2 \equiv -1 \pmod{s^2}$ has exactly $2^{\omega(s)}$ roots. Counting complete blocks in the 23 allowed residue classes gives

$$\left| A_s - \theta_q N \frac{2^{\omega(s)}}{s^2} \right| \leq 3 \cdot 2^{\omega(s)},$$

where A_s counts $x \in U_q$ with $s^2 \mid x^2 + 1$.

Every such s is coprime to both M_i . If $C_{M_i}(y) = \#\{n \leq y : (n, M_i) = 1\}$, inclusion–exclusion gives $C_{M_i}(y) \leq \eta_i y + 2^{\omega(M_i)}$. The hyperbola decomposition therefore gives

$$\begin{aligned} \sum_{\substack{s \leq R \\ (s, M_i) = 1}} d(s) &\leq 2 \sum_{\substack{a \leq \sqrt{R} \\ (a, M_i) = 1}} C_{M_i}(R/a) \\ &\leq 2\eta_i R \sum_{\substack{a \leq \sqrt{R} \\ (a, M_i) = 1}} \frac{1}{a} + e_i \sqrt{R} \leq K_i(R). \end{aligned}$$

The two first-block harmonic bounds are finite rational computations. The last inequality follows by separating the first block modulo M_i and bounding each later block by η_i/k . Hence the small- s rounding loss is at most $3K_i(R)$, while

$$\sum_{s > R} \frac{d(s)}{s^2} \leq \frac{2(\log R + 2)}{R}.$$

The remaining large-square tail is charged only once. Writing $x^2 + 1 = ds^2$, the negative Pell equation has at most $1 + \log_{2+\sqrt{3}} N$ positive solutions for each fixed d , so

$$\sum_{s > R} A_s \leq \frac{N^2 + 1}{R^2} \left(1 + \log_{2+\sqrt{3}} N\right).$$

Finally $|U_q| \leq \theta_q N + 23$. Combining the bounds and using $r_N \leq R \leq r_N + 1$ proves (6). Notice that $r_N \geq M_0^2$ already for $N \geq 7.8 \cdot 10^{10}$, whereas $r_N \geq M_1^2$ for $N \geq 2.274 \cdot 10^{13}$. Elementary differentiation proves the asserted monotonicity on the displayed ranges; direct endpoint evaluation gives the constants. $\square$

For the last refinement define

$$V_8 = \{x \leq N : x \not\equiv 7, 18 \pmod{25}, \quad (x \text{ odd or } 8 \mid x)\}$$

and, whenever $r_N \geq M_i^2$,

$$\begin{aligned} \Delta_8^{(i)}(N) &= \frac{115}{N} + \frac{6K_i(r_N + 1)}{N} + \frac{23(\log(r_N + 1) + 2)}{20r_N} \\ &\quad + \frac{N^2 + 1}{Nr_N^2} \left(1 + \log_{2+\sqrt{3}} N\right). \end{aligned} \quad (7)$$

Lemma 4.3 (Two-adic split of the even non-base branch). *Let $3 \cdot 10^{13} \leq N \leq 6 \cdot 10^{13}$ and let b be even. In the two $q_0 = 25$ progression estimates obtained from b , one may use*

$$H = 2048 \quad \text{if } \nu_2(b) = 1, \quad H = 4096 \quad \text{if } \nu_2(b) = 2, \quad H = 8192 \quad \text{if } \nu_2(b) \geq 3.$$

Moreover

$$\#\{x \in V_8 : \mu(x^2 + 1) = 0\} \leq N \left(\frac{23}{40} C_{\text{quad}} + \Delta_8^{(1)}(N) \right),$$

and at the left endpoint $\Delta_8^{(1)}(N) < 0.000667629$.

Proof. After removal of the redundant factor 5, the effective root-count modulus is $5b$. On the stated interval it is at most $3 \cdot 10^{14}$. If $\nu_2(b) = 1$, a root count greater than 2048 would require a modulus at least $2P_{12} = 304250263527210$; if $\nu_2(b) = 2$, a count greater than 4096 would require at least $4P_{12} = 608500527054420$. If $8 \mid b$, the existing 8192 bound applies. This proves the progression assertions.

For the diagonal assertion, the odd non-base classes have density $23/50$ and the classes divisible by 8 have density $23/200$. For each root modulo s^2 , counting the two parts successively modulo 2, 8, and 25 gives total discrepancy at most 6. Their union has 115 classes modulo 200, so the proof of Lemma 4.2, with small-modulus loss $6K_1(R)$ and the Pell tail still paid once, gives (7) with $i = 1$. Its terms decrease, and direct substitution gives the stated endpoint value. $\square$

Corollary 4.4 (Low-range two-adic diagonal). *For $N \geq 7.8 \cdot 10^{10}$,*

$$\#\{x \in V_8 : \mu(x^2 + 1) = 0\} \leq N \left(\frac{23}{40} C_{\text{quad}} + \Delta_8^{(0)}(N) \right).$$

The envelope decreases, and at the left endpoint

$$\Delta_8^{(0)}(N) < 0.004510951.$$

Proof. Repeat the diagonal part of Lemma 4.3 with M_0 in place of M_1 . Here $r_N \geq M_0^2$ throughout the stated range, so the same 115 residue-class endpoint loss, $6K_0(R)$ small-square loss, and single Pell tail give (7) with $i = 0$. Elementary differentiation shows that every term decreases; direct substitution at $N = 7.8 \cdot 10^{10}$ gives the displayed value. $\square$

For $\varepsilon \in \{0, 1, 2, 3\}$ put

$$W_\varepsilon = \{x \in U_{25} : x \not\equiv \varepsilon \pmod{4}\}$$

and define

$$\begin{aligned} \Delta_W(N) &= \frac{69}{N} + \frac{6K_0(r_N + 1)}{N} + \frac{69(\log(r_N + 1) + 2)}{50r_N} \\ &\quad + \frac{N^2 + 1}{Nr_N^2} \left(1 + \log_{2+\sqrt{3}} N\right). \end{aligned} \quad (8)$$

Lemma 4.5 (Three-mod-4-class diagonal). *For $N \geq 7.8 \cdot 10^{10}$ and every choice of ε ,*

$$\#\{x \in W_\varepsilon : \mu(x^2 + 1) = 0\} \leq N \left(\frac{69}{100} C_{\text{quad}} + \Delta_W(N) \right).$$

The envelope decreases, and at the left endpoint

$$\Delta_W(N) < 0.004511053.$$

Proof. For every squarefree small-square modulus s in the proof of Lemma 4.2, its root count on W_ε is the difference of its root counts on U_{25} and $U_{100,\varepsilon}$. Subtracting their main terms gives density $69/100$, while the two discrepancies give at most $6 \cdot 2^{\omega(s)}$. Also W_ε is a union of 69 classes modulo 100. Thus the same hyperbola calculation has endpoint loss 69, small-square loss $6K_0(R)$, harmonic-tail coefficient $69/50$, and one common Pell tail. These are exactly the four terms in (8). Their monotonicity and the displayed endpoint value follow by differentiation and direct substitution. $\square$

The final low-range refinement uses the stronger coprimality modulus before a complete residue block is available. Put

$$M_* = 43890, \quad \eta_* = \frac{288}{1463}, \quad h_* := \sum_{\substack{1 \leq a \leq 7194 \\ (a, M_*) = 1}} \frac{1}{a} < 2.329039$$

and

$$K_*(R) = 2\eta_* R h_* + 128\sqrt{R}.$$

The displayed bound for h_* is an exact finite rational summation. On $6.7 \cdot 10^{10} \leq N \leq 1.001 \cdot 10^{11}$ define

$$\begin{aligned} \tilde{\Delta}_q(N) &= \Delta_q^{(0)}(N) - \frac{3K_0(r_N + 1)}{N} + \frac{3K_*(r_N + 1)}{N} \quad (q = 25, 50, 100), \\ \tilde{\Delta}_W(N) &= \Delta_W(N) - \frac{6K_0(r_N + 1)}{N} + \frac{6K_*(r_N + 1)}{N}, \\ \tilde{\Delta}_8(N) &= \Delta_8^{(0)}(N) - \frac{6K_0(r_N + 1)}{N} + \frac{6K_*(r_N + 1)}{N}. \end{aligned}$$

Lemma 4.6 (Short-block diagonal refinement). *Throughout this interval the conclusions of Lemma 4.2, Corollary 4.4, and Lemma 4.5 hold with their respective error terms replaced by $\tilde{\Delta}_q$, $\tilde{\Delta}_8$, and $\tilde{\Delta}_W$. These envelopes decrease, and at $N = 6.7 \cdot 10^{10}$,*

$$\begin{aligned} \tilde{\Delta}_{25} &< 0.002514345, & \tilde{\Delta}_{50} &< 0.002513892, & \tilde{\Delta}_{100} &< 0.002513665, \\ \tilde{\Delta}_W &< 0.004175714, & \tilde{\Delta}_8 &< 0.004175602. \end{aligned}$$

Proof. For $R = r_N + 1$ on the stated interval, direct endpoint evaluation gives $\sqrt{R} < 7194$. Inclusion–exclusion gives

$$C_{M_*}(y) = \#\{n \leq y : (n, M_*) = 1\} \leq \eta_* y + 64.$$

The hyperbola step in Lemma 4.2 therefore yields, without requiring $R \geq M_*^2$,

$$\sum_{\substack{s \leq R \\ (s, M_*)=1}} d(s) \leq 2\eta_* R \sum_{\substack{a \leq \sqrt{R} \\ (a, M_*)=1}} \frac{1}{a} + 128\sqrt{R} \leq K_*(R).$$

Every squarefree modulus s occurring in the three diagonal arguments is coprime to M_* , because its prime factors are 1 (mod 4), whereas the added prime is 19. Thus only the displayed small-square rounding term changes; all main terms, harmonic tails, and the single Pell tail remain unchanged. This gives the three tilded formulas. Each normalized term decreases by elementary differentiation, and direct substitution gives the endpoint bounds. $\square$

Theorem 4.7 (Explicit large range). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3 \cdot 10^{13}$.*

Proof. If A is contained in A_7 or A_{18} , the result is immediate. Otherwise the exhaustive parity decomposition gives the following four normalized upper bounds, in order: A^* has an even element; A^* is odd and the base part is odd; A^* is odd and a base class has an even element; and $A^* = \emptyset$ with both base classes nonempty:

$$\begin{aligned} \frac{23}{25}C_{\text{quad}} + \frac{2}{25}C_{2,5} &< 0.0376115, \\ \frac{23}{50}C_{\text{quad}} + \frac{1}{50} + \frac{1}{50}C_{2,5} &< 0.0356927, \\ \frac{23}{50}C_{\text{quad}} + \frac{1}{25}C_5 + \frac{1}{25}C_{2,5} &< 0.0334755, \\ \frac{2}{25}C_5 &< 0.0293395. \end{aligned}$$

First let $3 \cdot 10^{13} \leq N < 6 \cdot 10^{13}$. In the last three cases, using $H = 8192$ in the progression envelope gives at the left endpoint the respective bounds

$$0.038548, \quad 0.036331, \quad 0.031792,$$

and all three decrease thereafter. In the first case choose an even $b \in A^*$. If $\nu_2(b) = 1$, Lemma 4.3 with the unrestricted diagonal estimate gives an endpoint bound below 0.039559; if $\nu_2(b) = 2$, it gives a bound below 0.039961. If neither type of element exists, every even element of A^* is divisible by 8, so $A^* \subseteq V_8$. The restricted diagonal estimate and $H = 8192$ then give

$$\frac{23}{40}C_{\text{quad}} + \frac{2}{25}C_{2,5} + \Delta_8^{(1)}(N) + 2\mathcal{E}_{25,8192}(N) < 0.031297$$

at the left endpoint, again with decreasing right side. Thus the entire interval $3 \cdot 10^{13} \leq N < 6 \cdot 10^{13}$ is closed.

It remains to prove the result for $N \geq 6 \cdot 10^{13}$. Indeed, in the first case an even member of A^* sieves both base classes. In the second, the four odd base progressions modulo 100 split into two

on which $4 \mid ab + 1$ automatically and two sieved progressions. In the third, an even base member sieves the opposite class and an odd member of A^* sieves its own base class. In the fourth, one member of each base class sieves the other. Lemmas 4.1 and 4.2 therefore give

$$\begin{aligned} U_1(N) &= 0.0376115 + \Delta_{25}^{(1)}(N) + 2\mathcal{E}_{25, H_{25}(N)}(N), \\ U_2(N) &= 0.0356927 + \Delta_{50}^{(1)}(N) + 2\mathcal{E}_{100, H_{100}(N)}(N) + 2/N, \\ U_3(N) &= 0.0334755 + \Delta_{50}^{(1)}(N) + 2\mathcal{E}_{25, H_{25}(N)}(N), \\ U_4(N) &= 0.0293395 + 2\mathcal{E}_{25, H_{25}(N)}(N). \end{aligned}$$

The only increases can occur at N_{100} and N_{25} . Evaluating the right-hand envelopes at the initial point and immediately after each jump gives

N	U_1	U_2	U_3	U_4
$6 \cdot 10^{13}$	0.0399044	0.0379855	0.0357684	0.0312858
N_{100}	0.0398938	0.0384785	0.0357578	0.0312767
N_{25}	0.0393740	0.0374551	0.0352380	0.0308770

All entries are strictly below $1/25$. Between rows they decrease. For $N \geq 1.6 \cdot 10^{15}$, use the original rounded progression errors $4.7N^{-1/4}$ and $5.3N^{-1/4}$ together with the decreasing aggregate diagonal bound; at the endpoint the four bounds are respectively below

$$0.039224, \quad 0.037495, \quad 0.035088, \quad 0.030827.$$

Thus every case is closed. The Hall statement follows from Theorem 2.2. $\square$

Corollary 4.8 (Parity pruning below the large close). *Let $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$. If a compatible outside set B violates the Hall inequality, then $B \setminus C_N^{18}$ contains an even element.*

Proof. Put $A = B \cup T_N(B)$ as in Theorem 2.2; then $A^* = B \setminus C_N^{18}$. If $A^* = \emptyset$, the pure opposite-base matching, Corollary 5.11, closes the Hall inequality. Suppose A^* is nonempty and odd. If the base part $A_7 \cup A_{18}$ is odd, use the low-range diagonal envelope

$$0.0356927 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{100, 8192}(N) + 2/N < 0.039999984$$

at $N = 8.858 \cdot 10^{12}$; it remains decreasing throughout the stated interval. If the base part has an even element, the corresponding bound

$$0.0334755 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{25, 8192}(N) < 0.037783$$

holds at $N = 8.858 \cdot 10^{12}$ and also decreases. Either alternative would give $|A| < N/25$, contradicting the assumed Hall defect. Hence A^* , and therefore $B \setminus C_N^{18}$, contains an even element. $\square$

Corollary 4.9 (Quantitative even-pivot pruning). *Let $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$, and suppose that a compatible outside set B violates the Hall inequality. Put*

$$E = \{b \in B \setminus C_N^{18} : 2 \mid b\}, \quad e = |E|,$$

and define

$$U_{\text{odd}}(N) = 0.0356927 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{100, 8192}(N) + 2/N.$$

Then

$$e > \left(\frac{1}{25} - U_{\text{odd}}(N) \right) N - \frac{7}{25} > 2.$$

Thus there are always two distinct even elements of $B \setminus C_N^{18}$. If $N \geq 10^{13}$, the stronger bound

$$e > \frac{21N}{125000} - \frac{7}{25}. \quad (9)$$

holds, and there are distinct even $b, c \in B \setminus C_N^{18}$ with

$$1 \leq |b - c| \leq 5952.$$

For this pair the prime 2 cannot occur in the same-prime part and

$$|S_N(b, c)| \leq \frac{58N}{11025} + 2. \quad (10)$$

The original Hall inequality follows if

$$|T_N(B) \cap D_N(b, c)| \leq \frac{383N}{11025} - |B| - \frac{57}{25}. \quad (11)$$

If moreover $2.25 \cdot 10^{13} \leq N < 3 \cdot 10^{13}$, the pair may be chosen with $|b - c| \leq 880$. Then at most one same-prime witness occurs,

$$|S_N(b, c)| \leq \frac{N}{225} + 1, \quad (12)$$

and it is sufficient that

$$|T_N(B) \cap D_N(b, c)| \leq \frac{8N}{225} - |B| - \frac{32}{25}. \quad (13)$$

Thus every residual defect in this range is unconditionally excluded from the one-even-pivot and 2-adic same-prime branches.

Proof. Put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$, and remove the set E from A . The remaining non-base part is odd. If it is nonempty, the larger of the two low-range odd envelopes in the preceding proof is at most $U_{\text{odd}}(N)N$. If it is empty, retain any $b \in E$, whose existence follows from Corollary 4.8. The two off-diagonal progression estimates from b give instead

$$|A \setminus E| \leq N \left(\frac{2}{25} C_{2,5} + 2\mathcal{E}_{25,8192}(N) \right) < 0.016136N$$

at $N = 8.858 \cdot 10^{12}$, with a decreasing right side. Hence in all cases $|A \setminus E| \leq U_{\text{odd}}(N)N$. A Hall defect has $|A| > |C_N| \geq N/25 - 7/25$, so subtraction proves the first displayed bound. At the left endpoint $U_{\text{odd}}(N) < 0.039999984$, so its right side exceeds $1.6 \cdot 10^{-8}N - 7/25 > 2$. At $N = 10^{13}$ one has $U_{\text{odd}}(N) < 0.039832$, and monotonicity proves (9) on the remaining range.

List the e even elements increasingly. Since

$$\frac{21}{125000} - \frac{1}{5953} = \frac{13}{744125000}$$

and $N \geq 10^{13}$, equation (9) gives $e - 1 > (N - 1)/5953$. A least consecutive gap is therefore less than 5953; being even, it is at most 5952.

For an even pivot, $ab + 1$ is odd, so 2 cannot witness a square divisor. Every same-prime witness also differs from 5 and has $p^2 \mid |b - c|$. There are at most two such primes, because $(3 \cdot 7 \cdot 11)^2 = 53361 > 5952$. The two largest residue-class budgets are therefore those for $p = 3, 7$, and

$$|S_N(b, c)| \leq \frac{N}{25} \left(\frac{1}{9} + \frac{1}{49} \right) + 2 = \frac{58N}{11025} + 2.$$

Combining this with (11) gives

$$|B| + |T_N(B)| \leq |B| + \frac{58N}{11025} + 2 + \frac{383N}{11025} - |B| - \frac{57}{25} = \frac{N}{25} - \frac{7}{25} \leq |C_N|,$$

which is the original Hall inequality.

At $N = 2.25 \cdot 10^{13}$, direct substitution in the decreasing low-range odd envelope gives

$$0.04 - \left(0.0356927 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{100,8192}(N) + 2/N \right) > \frac{1}{881} + 4 \cdot 10^{-6}.$$

The preceding argument therefore gives $e - 1 > (N - 1)/881$ throughout the final subinterval. The least even gap is less than 881, hence at most 880. Two distinct same-prime witnesses would force $(3 \cdot 7)^2 = 441$ to divide this even gap, and consequently would force the gap to be at least 882, a contradiction. Thus only $p = 3$ need be charged, which proves (12). Subtracting its bound from $|C_N| \geq N/25 - 7/25$ gives (13). $\square$

Lemma 4.10 (Finite two-pivot sieve on the base classes). *Let $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$, and let $b, c \leq N$ be distinct even integers, neither congruent to 7 or 18 (mod 25). Then*

$$\#\{a \leq N : a \equiv 7 \text{ or } 18 \pmod{25}, \mu(ab + 1) = \mu(ac + 1) = 0\} < 0.013992N.$$

If in addition $N \geq 10^{13}$ and $|b - c| \leq 5952$, the right side improves to $0.012132N$.

Proof. Let

$$\mathcal{P} = \{3, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}.$$

For $p \in \mathcal{P}$, each of $p^2 \mid ab + 1$ and $p^2 \mid ac + 1$ specifies at most one residue of a modulo p^2 . When both residues exist, they coincide exactly when $p^2 \mid b - c$.

Fix one of the two classes $a \equiv 7, 18 \pmod{25}$. Inclusion–exclusion for the event that some $p \in \mathcal{P}$ divides $ab + 1$ to second order, the analogous event for $ac + 1$, and their union gives the local intersection density

$$I_J = 1 - 2 \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2} \right) + \prod_{p \in J} \left(1 - \frac{1}{p^2} \right) \prod_{p \in \mathcal{P} \setminus J} \left(1 - \frac{2}{p^2} \right),$$

where $J = \{p \in \mathcal{P} : p^2 \mid b - c\}$. In general it is largest when $J = \mathcal{P}$, and direct rational multiplication gives

$$I_J \leq I_{\mathcal{P}} = 1 - \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2} \right) < 0.152377. \quad (14)$$

If $|b - c| \leq 5952$, at most two primes lie in J , because $(3 \cdot 7 \cdot 11)^2 > 5952$. In that case the worst choice is $J = \{3, 7\}$ and $I_J < 0.129757$. Every nonempty inclusion–exclusion intersection is one CRT class after $a \pmod{25}$ is fixed. The total rounding discrepancy is at most

$$\epsilon = 2(2^{13} - 1) + (3^{13} - 1) = 1610704.$$

Thus the small-prime common failures in both base classes contribute at most

$$\frac{2I_J}{25}N + 2\epsilon. \quad (15)$$

It remains to charge a witness prime larger than 47 in at least one of the two linear forms. Put $Y = \lfloor N/333 \rfloor$. The finite computation

$$\sum_{47 < p \leq 1000} \frac{1}{p^2} + \sum_{n > 1000} \frac{1}{n^2} < 0.004751 =: \tau$$

gives $\sum_{p > 47} p^{-2} < \tau$. For $47 < p \leq Y$, a fixed prime contributes at most $N/(25p^2) + 1$ in a fixed base class. Hence one form in one base class contributes at most

$$\frac{\tau N}{25} + \pi(Y)$$

in this intermediate range.

For $p > Y$, use the tail calculation in Proposition 1 of Sothanaphan [2]. After the same removal of one redundant factor 5 as in Lemma 4.1, the root count is at most 8192 because $5N < Q_*$. If $X \leq N/25 + 1$ is the length of the base progression and d_1 is its transformed modulus, then

$$\sum_{r > Y} A_r \leq 8192 \left(\frac{2x}{d_1 Y} + \frac{x}{Y^2} \right) \leq L(N),$$

where

$$L(N) = 8192 \left(\frac{2(N/25 + 2)}{Y} + \frac{N^2 + 25N + 1}{Y^2} \right).$$

This bounds, in particular, the prime-square tail. There are two forms and two base classes, so (15) and the union bound give

$$\frac{|G_N(b, c)|}{N} \leq \frac{2I_J}{25} + \frac{2\epsilon}{N} + \frac{4\tau}{25} + \frac{4\pi(Y)}{N} + \frac{4L(N)}{N}, \quad (16)$$

where $G_N(b, c)$ denotes the set in the statement.

Rosser and Schoenfeld's explicit estimate $\pi(y) < 1.25506y/\log y$ for $y > 1$ [3] applies. Since $N/334 \leq Y \leq N/333$, all nonconstant terms in the resulting upper envelope decrease for $N \geq 8.858 \cdot 10^{12}$. At the left endpoint its five terms, using $I_J < 0.152377$, are respectively less than

$$0.012190160, \quad 0.000000364, \quad 0.000760160, \quad 0.000628127, \quad 0.000412773.$$

Their sum is less than 0.013992. Under the additional gap hypothesis, evaluate instead at $N = 10^{13}$ with $I_J < 0.129757$; the five terms are respectively less than

$$0.01038056, \quad 0.000000323, \quad 0.000760160, \quad 0.000624970, \quad 0.000365635.$$

and their sum is less than 0.012132. This proves both assertions. $\square$

Theorem 4.11 (Improved explicit large range). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 8.858 \cdot 10^{12}$.*

Proof. Theorem 4.7 handles $N \geq 3 \cdot 10^{13}$. Suppose instead that $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$ and that B is a Hall defect. Put $A = B \cup T_N(B)$. Corollary 4.9 supplies distinct even $b, c \in B \setminus C_N^{18}$.

Every $x \in A^*$ has $\mu(x^2 + 1) = 0$, so the repaired low-range aggregate diagonal estimate gives

$$|A^*| \leq N \left(\frac{23}{25} C_{\text{quad}} + \Delta_{25}^{(0)}(N) \right) < 0.025784N.$$

Every member of $A_7 \cup A_{18}$ is a common non-neighbour of b, c . Lemma 4.10 therefore gives

$$|A| < (0.025784 + 0.013992)N = 0.039776N < \frac{N}{25} - \frac{7}{25} \leq |C_N|,$$

contradicting the Hall defect. Thus the middle interval is closed, and the Hall equivalence finishes the proof. $\square$

5 A direct singleton Hall close

The full large- N argument spends density on all parts of an admissible set. For a singleton Hall defect we only need one squarefree neighbour, and the same explicit progression estimate becomes effective much earlier.

We use the following all-parameter form of the Möbius splitting estimate proved in Proposition 1 of Sothanaphan [2]. If $x \geq q \geq 1$, $(u, q) = 1$, and $D \geq 1$ is an integer, then

$$S_2(x; q, u) \geq \frac{x}{q} \prod_{p \nmid q} \left(1 - \frac{1}{p^2} \right) - D - \frac{x}{qD} - \lambda q^{1/4} \left(\frac{2x}{qD} + \frac{x}{D^2} \right), \quad (17)$$

where $S_2(x; q, u)$ counts squarefree positive integers at most x in the class $u \pmod{q}$ and

$$\lambda = \frac{128}{(8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} < 6.876.$$

For $N \geq 2 \cdot 10^9$ define

$$\begin{aligned} \mathcal{L}(N) = & 0.6079 \left(\frac{N}{25} - 2 \right) - (3.2N^{3/4} + 1) - \frac{N/25 + 1}{3.2N^{3/4}} \\ & - 6.876\sqrt{5}N^{1/4} \left(\frac{2(N/25 + 1)}{3.2N^{3/4}} + \frac{N^2 + 25N}{(3.2N^{3/4})^2} \right). \end{aligned} \quad (18)$$

Theorem 5.1 (Uniform one-pivot Hall expansion). *Let $N \geq 2 \cdot 10^9$, and let $b \in [1, N] \setminus C_N$. Then there is an $a \in C_N$ for which $ab + 1$ is squarefree. More precisely,*

$$|\Gamma_N(\{b\})| \geq \mathcal{L}(N) > 0.00207N,$$

and $\mathcal{L}(2 \cdot 10^9) > 4.14 \cdot 10^6$. Consequently every nonempty compatible outside set B with $|B| \leq \mathcal{L}(N)$ satisfies the Hall inequality (2). This includes all singleton defects.

Proof. Write $a = 7 + 25n$ and put

$$X = |C_N|, \quad g = (7b + 1, 25b), \quad q = \frac{25b}{g}, \quad c = \frac{7b + 1}{g}.$$

Since $(b, 7b + 1) = 1$, one has $g \mid 25$. The exclusion $b \not\equiv 7 \pmod{25}$ gives $g \in \{1, 5\}$, $(c, q) = 1$, and $g \mid q$ but $g \nmid c$. Therefore g is squarefree and

$$(7 + 25n)b + 1 = g(qn + c)$$

is squarefree exactly when $qn + c$ is squarefree.

The X values $qn + c$, $0 \leq n < X$, are precisely the positive integers in the class $c \pmod{q}$ up to

$$x = q(X - 1) + c.$$

Here $c \leq q$, $q \leq 25N$, $x/q \leq X$, and

$$\frac{x}{q} \geq X - 1, \quad \frac{N}{25} - 1 \leq X \leq \frac{N}{25} + 1, \quad x \leq N^2 + 25N.$$

Take $D = \lceil 3.2N^{3/4} \rceil$ in (17). Since

$$\prod_{p \nmid q} \left(1 - \frac{1}{p^2}\right) \geq \prod_p \left(1 - \frac{1}{p^2}\right) = \frac{6}{\pi^2} > 0.6079$$

and $q^{1/4} \leq \sqrt{5}N^{1/4}$, the number of squarefree values is at least $\mathcal{L}(N)$. After division by N , every subtracted term is decreasing for $N \geq 2 \cdot 10^9$, while the positive main term is increasing. Using $\sqrt{5} < 2.237$, at that endpoint this lower bound exceeds $4.14 \cdot 10^6$ and its quotient by N exceeds 0.00207. Thus a squarefree neighbour exists, and the normalized lower bound remains above 0.00207 thereafter. For a nonempty B , choose $b \in B$. Then $\Gamma_N(\{b\}) \subseteq \Gamma_N(B)$, so $|\Gamma_N(B)| \geq \mathcal{L}(N) \geq |B|$ whenever the displayed size hypothesis holds. This is precisely the Hall inequality. $\square$

For the matching range we can reuse the primorial progression error instead of the coarser all-parameter singleton bound. Put

$$N_{4096} = \frac{8P_{11}}{5} = 5936590507848, \quad H_{\deg}(N) = \begin{cases} 4096, & N < N_{4096}, \\ 8192, & N \geq N_{4096}, \end{cases}$$

and set

$$\kappa_5 = \frac{12769}{20164}.$$

For $1.8263 \cdot 10^{11} \leq N < N_{25}$, put

$$\mathcal{M}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25, H_{\deg}(N)}(N).$$

Proposition 5.2 (Primorial one-pivot degree). *On this interval every outside vertex has at least $\mathcal{M}(N)$ squarefree neighbours in C_N , and every vertex of C_N has at least the same number in C_N^{18} . Moreover $\mathcal{M}(N)/N$ is increasing on each of the two intervals separated by N_{4096} , and*

$$\mathcal{M}(1.8263 \cdot 10^{11}) > \frac{N}{50} + 2.8 \cdot 10^4.$$

Proof. For an outside pivot, Corollary 1 of Sothanaphan's progression estimate gives the bad-value upper bound used in (3). The effective root modulus is at most $5N$. The least moduli with root count greater than 4096 and 8192 are $8P_{11}$ and Q_* , respectively, so Lemma 4.1 allows the displayed $H_{\deg}$. Also

$$\prod_{(p,25b)=1} \left(1 - \frac{1}{p^2}\right) \geq \prod_{p \neq 5} \left(1 - \frac{1}{p^2}\right) = \frac{25}{4\pi^2} > \kappa_5.$$

Indeed the classical bound $\pi < 355/113$ gives

$$\frac{25}{4\pi^2} > \frac{25}{4} \left(\frac{113}{355}\right)^2 = \frac{12769}{20164} = \kappa_5.$$

Since $|C_N| \geq N/25 - 7/25$, subtraction gives the claimed lower degree. The reverse 7–18 calculation is identical. Every nonconstant term of $\mathcal{E}_{25,H_{\deg}}$ decreases while $H_{\deg}$ is fixed. Direct evaluation gives the displayed lower-end margin and $\mathcal{M}(N_{4096})/N_{4096} > 0.0222$ immediately after the sole jump. This proves both the asserted piecewise monotonicity and the strict half-class lower bound throughout the interval. $\square$

For $4.5654 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}$ define the correlated degree envelope

$$\mathcal{M}_1(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,1024}(N).$$

Proposition 5.3 (Root-product correlated degree). *Throughout this interval every outside vertex has at least $\mathcal{M}_1(N)$ squarefree neighbours in C_N , and the same reverse degree holds from C_N to C_N^{18} . Moreover $\mathcal{M}_1(N)/N$ is increasing and*

$$\mathcal{M}_1(4.5654 \cdot 10^{10}) > \frac{N}{50} + 800.$$

Proof. Let $m = 2^\nu m_0 \leq 5N$ be the effective root modulus. If $\rho(m) \leq 1024$, the proof of Proposition 5.2 applies directly with $H = 1024$. Suppose instead that $\rho(m) > 1024$. Since $5N < P_{11}$ throughout the present interval, the cases $\nu \leq 1$ are impossible: they require at least eleven odd prime factors. Removing a factor 5 did not change the 2-adic valuation, so $\nu \geq 2$ implies $4 \mid b$. The Euler product then omits $p = 2$ as well as $p = 5$, and is at least $4\kappa_5/3$; meanwhile $\rho(m) \leq 4096$, since the least modulus with more than 4096 roots is $8P_{11} > 5N$. Thus the $H = 4096$ lower degree is at least the asserted $H = 1024$ lower degree provided

$$\mathcal{E}_{25,4096}(N) - \mathcal{E}_{25,1024}(N) \leq \frac{\kappa_5}{75}.$$

The left side decreases; at $N = 4.5654 \cdot 10^{10}$ it is below 0.003132, whereas $\kappa_5/75 > 0.008443$. This proves the correlated bound in both directions. The monotonicity and endpoint margin follow directly from the formula for $\mathcal{E}_{25,1024}$. $\square$

For $2.2827 \cdot 10^{10} \leq N < 2.47 \cdot 10^{11}$ put

$$\mathcal{M}_{1/2}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N).$$

Proposition 5.4 (Second root–Euler correlated degree). *Throughout this interval every outside vertex has at least $\mathcal{M}_{1/2}(N)$ squarefree neighbours in C_N , and the same reverse degree holds from C_N to C_N^{18} . The quotient $\mathcal{M}_{1/2}(N)/N$ increases, and*

$$\mathcal{M}_{1/2}(2.2827 \cdot 10^{10}) > \frac{N}{50} + 411.$$

Proof. Let $m = 2^\nu m_0 \leq 5N$ be the effective root modulus. If $\rho(m) \leq 512$, use the progression estimate directly with $H = 512$. Suppose $\rho(m) > 512$ and $\nu \leq 1$. Then m_0 has at least ten distinct odd prime factors. At most one is the modulus prime 5, so the pivot is divisible by at least nine distinct odd primes other than 5. If 3 did not divide the pivot, then

$$b \geq 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37 = 247357937827 > N,$$

a contradiction. Hence the Euler product omits 3 and is at least $9\kappa_5/8$. Also $\rho(m) \leq 1024$, because $5N < P_{11}$, and at the lower endpoint

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} < 0.001386 < \frac{\kappa_5}{200}.$$

Thus the missing-3 main-term gain pays the larger root error.

If $\nu \geq 2$, then $4 \mid b$, the Euler product omits 2, and $\rho(m) \leq 4096$. At the same endpoint

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} < 0.005332 < \frac{\kappa_5}{75},$$

so the missing-2 gain pays this error. Both differences decrease. The reverse progression has the identical alternatives. This proves the degree bounds; monotonicity and the endpoint margin follow by direct evaluation. $\square$

Put

$$\beta_8 := 0.00097181149.$$

For $1.382 \cdot 10^{10} \leq N \leq 2.2827 \cdot 10^{10}$ define

$$\mathcal{M}_{1/4}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta_8 N.$$

Proposition 5.5 (Eight-prime root–Euler correlated degree). *Throughout this interval every outside vertex has at least $\mathcal{M}_{1/4}(N)$ squarefree neighbours in C_N , and the same reverse degree holds from C_N to C_N^{18} . The quotient $\mathcal{M}_{1/4}(N)/N$ increases, and*

$$\mathcal{M}_{1/4}(1.382 \cdot 10^{10}) > \frac{N}{50} + 16356.$$

Proof. If the effective root count is at most 256, use $H = 256$. Direct endpoint evaluation gives

$$\mathcal{E}_{25,512} - \mathcal{E}_{25,256} > 0.001099 > \beta_8$$

throughout the interval.

Suppose next that the root count exceeds 256 and the effective modulus has 2-adic valuation at most one. It has at least nine distinct odd prime factors. At most one is the modulus prime

5, so the pivot is divisible by at least eight distinct non-5 odd primes. Exact ordered-prime enumeration under product at most $2.2827 \cdot 10^{10}$ gives the unique least Euler gain at

$$\{7, 13, 17, 19, 23, 29, 31, 37\}, \quad \prod p = 22487085257,$$

and

$$\prod_p \left(1 - \frac{1}{p^2}\right)^{-1} - 1 > 0.038365586453.$$

Consequently its normalized main-term gain is greater than

$$\frac{\kappa_5}{25}(0.038365586453) > 0.000971811492 > \beta_8.$$

If the 2-adic valuation is at least two, the missing-2 gain $\kappa_5/75$ is larger still. Thus every root count at most 512 pays the claimed improved degree.

For the remaining higher-root strata, at the lower endpoint

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta_8 < 0.002610 < \frac{\kappa_5}{200},$$

and

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta_8 < 0.007275 < \frac{\kappa_5}{75}.$$

The forced missing-3 and missing-2 gains from the proof of Proposition 5.4 therefore continue to pay those strata. The reverse progression is identical. Monotonicity and the endpoint margin follow from the displayed formula. $\square$

Put

$$\beta'_8 := 0.00111752951, \quad \mathcal{M}_{1/8}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta'_8 N.$$

Proposition 5.6 (Second eight-prime root–Euler degree). *For $1.29 \cdot 10^{10} \leq N \leq 1.382 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{1/8}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{1/8}(1.29 \cdot 10^{10}) > \frac{N}{50} + 6728.$$

Proof. Repeat Proposition 5.5, but impose the current product bound $1.382 \cdot 10^{10}$. Complete ordered-prime enumeration now visits 828 nodes and is uniquely minimized by

$$\{7, 11, 13, 17, 23, 29, 31, 37\}, \quad \prod p = 13018838833.$$

Exact rational multiplication gives

$$\prod_p \left(1 - \frac{1}{p^2}\right)^{-1} - 1 = \frac{13286935886873651}{301166059585536000} > 0.044118304383,$$

and hence normalized gain greater than $0.001117529515 > \beta'_8$. The low-root saving $\mathcal{E}_{25,512} - \mathcal{E}_{25,256}$ exceeds 0.001299, while at the lower endpoint the two remaining comparisons are

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta'_8 < 0.002794 < \frac{\kappa_5}{200}$$

and

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta'_8 < 0.007568 < \frac{\kappa_5}{75}.$$

Thus every root stratum pays the asserted degree. Direct endpoint evaluation gives the margin. $\square$

Put

$$\beta_8'' := 0.00113583855, \quad \mathcal{M}_{1/16}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta_8'' N.$$

Proposition 5.7 (Third eight-prime root–Euler degree). *For $1.279 \cdot 10^{10} \leq N \leq 1.29 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{1/16}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{1/16}(1.279 \cdot 10^{10}) > \frac{N}{50} + 5090.$$

Proof. At product bound $1.29 \cdot 10^{10}$, complete enumeration is uniquely minimized by

$$\{7, 11, 13, 17, 19, 29, 31, 43\}, \quad \prod p = 12498697211.$$

Its exact relative Euler gain is

$$\frac{12438467965272539}{277389791723520000} > 0.044841116495,$$

which gives normalized gain $> 0.001135838556 > \beta_8''$. At the lower endpoint the low-root saving and the two higher-root comparisons remain strict:

$$\begin{aligned} \mathcal{E}_{25,512} - \mathcal{E}_{25,256} &> \beta_8'', \\ \mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta_8'' &< 0.002799 < \frac{\kappa_5}{200}, \\ \mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta_8'' &< 0.007586 < \frac{\kappa_5}{75}. \end{aligned}$$

Thus the same root-stratified degree proof applies, and endpoint evaluation gives the margin. $\square$

Put

$$\mathcal{M}_{\text{QR2}}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,128}(N).$$

Proposition 5.8 (Quadratic-residue root hierarchy). *For $7.67769 \cdot 10^9 \leq N \leq 1.15 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{\text{QR2}}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{\text{QR2}}(7.67769 \cdot 10^9) > \frac{N}{50} + 3853771.$$

Proof. The root count is at most 2048: more roots would require at least nine non-5 odd prime divisors, whose least product is

$$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 = 20056049013 > N.$$

If the root count is at most 128, use Lemma 4.1 directly. For the four remaining root strata, let k be the number of non-5 odd prime divisors. Root count in $(128, 256]$, $(256, 512]$, $(512, 1024]$, or $(1024, 2048]$ forces $k \geq 5, 6, 7, 8$, respectively.

Each stratum also forces enough small prime divisors. If the asserted number were absent, the least possible radical would already exceed the upper endpoint:

H	forced small primes	excluded least product	value
256	3 primes ≤ 1000	$3 \cdot 7 \cdot 1009 \cdot 1013 \cdot 1019$	21872281683
512	4 primes ≤ 360	$3 \cdot 7 \cdot 11 \cdot 367 \cdot 373 \cdot 379$	11984670159
1024	5 primes ≤ 160	$3 \cdot 7 \cdot 11 \cdot 13 \cdot 163 \cdot 167 \cdot 173$	14141826699
2048	6 primes ≤ 60	$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 61 \cdot 67 \cdot 71$	14813826027.

Choose $D = 20000000$. Then the complementary variable satisfies

$$m \leq \frac{N^2 + 25N + 1}{D^2} \leq 330626.$$

Intersect every possible collection of the forced quadratic-residue masks. The exact maxima are

H	mask cutoff/size	max $\#m$	H max $\#m$
256	1000/3	42697	10930432
512	360/4	21570	11043840
1024	160/5	10663	10918912
2048	60/6	4798	9826304.

Thus every higher-root tail is at most 11043840. The calculation is reproduced by `scripts/quadratic_residue_hierarchy_scan.py`.

Discarding every Euler gain, the progression proof gives at the new lower endpoint more than

$$\frac{\kappa_5(N - 43)}{25} - D - \frac{N/25 + 1}{D} - \frac{1}{5D} - 11043840 - 1 > \frac{N}{50} + 9933718.$$

This lower bound increases with N , whereas $\mathcal{M}_{\text{QR2}}(N) - N/50 < 1.277 \cdot 10^7$ throughout the interval. Hence every higher-root stratum also supplies the asserted degree. Direct evaluation gives the displayed endpoint margin. $\square$

Put

$$\mathcal{M}_{\text{QR}}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,256}(N).$$

Proposition 5.9 (Quadratic-residue screened degree). *For $1.15 \cdot 10^{10} \leq N \leq 1.2643 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{\text{QR}}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{\text{QR}}(1.15 \cdot 10^{10}) > \frac{N}{50} + 154290.$$

Proof. If the transformed root count is at most 256, this is Lemma 4.1 with $H = 256$. We replace the crude root payment only in the stratum $256 < \rho(d_1) \leq 512$ whose 2-adic exponent is at most one. There the effective modulus contains exactly eight non-5 odd prime divisors S . In the large-square tail of Proposition 1 of Sothanaphan, the complementary variable m and a root d satisfy

$$d^2 m \equiv c_1 \pmod{d_1}.$$

For every $p \in S$ one has $c_1 \equiv 1 \pmod{p}$; hence any contributing m must be a nonzero quadratic residue modulo every $p \in S$.

Choose $D = 8000000$. On the whole interval,

$$m \leq \frac{N^2 + 25N + 1}{D^2} \leq 2497585.$$

Complete ordered-prime enumeration at the upper endpoint gives 5700 feasible supports, involving 101 possible primes. Intersecting the exact quadratic-residue masks through 2497585 leaves at most 6024 values of m for every support. Since each has at most 512 roots, the complete tail is at most

$$512 \cdot 6024 = 3084288.$$

The scan is reproduced by `scripts/quadratic_residue_support_scan.py`.

We may ignore the Euler gain from all eight primes. If X is either base progression length, then $X - 1 \geq (N - 43)/25$, $X \leq N/25 + 1$, and the proof of the progression estimate now gives at least

$$\frac{\kappa_5(N - 43)}{25} - D - \frac{N/25 + 1}{D} - \frac{1}{5D} - 3084288 - 1 > \frac{N}{50} + 50214005$$

squarefree values at the lower endpoint. The left side minus $N/50$ increases, while $\mathcal{M}_{\text{QR}}(N) - N/50$ is below $2.26 \cdot 10^6$ on the interval. Thus this root stratum more than pays the asserted degree.

For root count above 512 with 2-adic exponent at most one, the forced missing-3 gain applies and, at the lower endpoint,

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,256} < 0.003124 < \frac{\kappa_5}{200}.$$

For 2-adic exponent at least two, the missing-2 gain applies and

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,256} < 0.008083 < \frac{\kappa_5}{75}.$$

Both differences decrease with N . This covers every root stratum in both directions. Direct endpoint substitution gives the stated margin. $\square$

Put

$$\beta_8''' := 0.00116052, \quad \mathcal{M}_{1/32}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta_8''' N.$$

Proposition 5.10 (Modulus-correlated eight-prime degree). *For $1.2643 \cdot 10^{10} \leq N \leq 1.279 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{1/32}(N)$ squarefree neighbours in C_N , with the same reverse degree. Throughout this interval the degree is strictly larger than half a base class, and at the lower endpoint*

$$\mathcal{M}_{1/32}(1.2643 \cdot 10^{10}) > \frac{N}{50} + 1442.$$

Proof. We retain the transformed modulus in the proof of Lemma 4.1. For $q_0 = 25$, root bound H , pivot b , and $X \leq N/25 + 1$, the last term before the uniform enlargement $d_1 \leq 25N$ is

$$\frac{Hd_1X}{D^2} \leq \frac{25HbX}{D^2}.$$

Consequently, with $H = 512$ and $d_{512} = 1024^{1/3}N^{2/3}$, the actual error is at most

$$N\mathcal{E}_{25,512}(N) - \frac{512 \cdot 25(N - b)(N/25 + 1)}{d_{512}^2}. \quad (19)$$

The reciprocal d_1 term is still bounded with $d_1 \geq 5$, so this is a one-sided strengthening of the already proved envelope.

Only the root stratum $256 < \rho(d_1) \leq 512$ with odd b needs the new correlation. In this stratum b has exactly eight distinct non-5 odd prime divisors. Write their set as S and their product as $r(S)$. Every prime divisor of b lies in $S \cup \{5\}$; hence

$$b = r(S)k, \quad k \leq N/r(S), \quad p \mid k \implies p \in S \cup \{5\}.$$

For fixed S , the Euler gain is

$$G(S) = \frac{\kappa_5}{25} \left(\prod_{p \in S} (1 - p^{-2})^{-1} - 1 \right),$$

whereas (19) is worst at the largest admissible k . Complete ordered-prime enumeration through the upper endpoint visits 5831 supports and 157 support/multiplier breakpoints. Direct differentiation shows that the degree for each fixed admissible pair (S, b) increases between consecutive breakpoints. The unique global minimum is at $N = 12643000000$, with

$$S = \{7, 11, 13, 17, 19, 29, 31, 43\}, \quad r(S) = b = 12498697211,$$

and exact Euler contribution

$$G(S) = \frac{158826797448565050491}{139832194007826432000000}.$$

Adding the retained-modulus saving gives

$$G(S) + \frac{512 \cdot 25(N - b)(N/25 + 1)}{d_{512}^2 N} > 0.001160529889 > \beta_8'''.$$

This finite calculation is reproduced by `scripts/modulus_correlated_eight_prime_scan.py`.

If the root count is at most 256, then at the lower endpoint

$$\mathcal{E}_{25,512} - \mathcal{E}_{25,256} > 0.00133914 > \beta_8'''.$$

If b is even, the missing-2 gain is already larger than the asserted bonus. The remaining higher-root strata satisfy

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta_8''' < 0.002848 < \frac{\kappa_5}{200},$$

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta_8''' < 0.007653 < \frac{\kappa_5}{75},$$

so the forced missing-3 and missing-2 gains pay them exactly as before. The reverse progression has the same alternatives. Endpoint substitution gives the displayed margin; the breakpoint enumeration and piecewise monotonicity preserve it on the full interval. $\square$

Define the degree function

$$\mathcal{D}(N) = \begin{cases} \mathcal{M}_{\text{QR}2}(N), & 7.67769 \cdot 10^9 \leq N < 1.15 \cdot 10^{10}, \\ \mathcal{M}_{\text{QR}}(N), & 1.15 \cdot 10^{10} \leq N < 1.2643 \cdot 10^{10}, \\ \mathcal{M}_{1/32}(N), & 1.2643 \cdot 10^{10} \leq N < 1.279 \cdot 10^{10}, \\ \mathcal{M}_{1/16}(N), & 1.279 \cdot 10^{10} \leq N < 1.29 \cdot 10^{10}, \\ \mathcal{M}_{1/8}(N), & 1.29 \cdot 10^{10} \leq N < 1.382 \cdot 10^{10}, \\ \mathcal{M}_{1/4}(N), & 1.382 \cdot 10^{10} \leq N < 2.2827 \cdot 10^{10}, \\ \mathcal{M}_{1/2}(N), & 2.2827 \cdot 10^{10} \leq N < 2.47 \cdot 10^{11}, \\ \mathcal{M}_1(N), & 2.47 \cdot 10^{11} \leq N < 3.4375 \cdot 10^{11}, \\ \mathcal{M}(N), & 3.4375 \cdot 10^{11} \leq N < N_{25}, \\ \mathcal{L}(N), & N \geq N_{25}. \end{cases}$$

Corollary 5.11 (Global matching of the opposite base class). *Let*

$$C_N^{18} = \{b \in [1, N] : b \equiv 18 \pmod{25}\}.$$

For every $N \geq 7.67769 \cdot 10^9$, the bipartite squarefree-edge graph between C_N^{18} and C_N has a matching that saturates C_N^{18} . Consequently every outside set $B \subseteq C_N^{18}$ satisfies the Hall inequality (2).

Proof. Below $1.15 \cdot 10^{10}$ use Proposition 5.8, from there to $1.2643 \cdot 10^{10}$ use Proposition 5.9, from there to $1.279 \cdot 10^{10}$ use Proposition 5.10, from there to $1.29 \cdot 10^{10}$ use Proposition 5.7, from there to $1.382 \cdot 10^{10}$ use Proposition 5.6, from there to $2.2827 \cdot 10^{10}$ use Proposition 5.5; from there to $2.47 \cdot 10^{11}$, Proposition 5.4 gives degree at least $\mathcal{D}(N)$ in both directions; from there to $3.4375 \cdot 10^{11}$ use Proposition 5.3, and from there to N_{25} use Proposition 5.2. Above that point, Theorem 5.1 gives degree at least $\mathcal{D}(N)$ from every vertex of C_N^{18} into C_N . Applying the same proof with the roles reversed and the progression $b \equiv 18 \pmod{25}$ gives the same lower bound from every vertex of C_N into C_N^{18} : in that calculation $(18a + 1, 25a) = 1$ for $a \equiv 7 \pmod{25}$, and all the inequalities used in Theorem 5.1 remain valid.

The quotient $\mathcal{L}(N)/N$ is increasing. Direct substitution at $N = 1.409 \cdot 10^{12}$ gives

$$\mathcal{L}(N) > \frac{N}{50} + 5.8 \cdot 10^5 > \frac{1}{2} \max\{|C_N^{18}|, |C_N|\}.$$

For the lower intervals, Propositions 5.8, 5.9, 5.10, 5.7, 5.6, 5.5, 5.4, 5.3, and 5.2, together with their endpoint and switch evaluations, give the same strict half-size inequality with $\mathcal{D}(N)$. Write $L = C_N^{18}$ and $R = C_N$. If $S \subseteq L$ is nonempty and $|S| \leq \mathcal{D}(N)$, the neighbourhood of any member of S already has size at least $|S|$. If $|S| > \mathcal{D}(N)$ and $|\Gamma(S)| < |S|$, then, because $|L| \leq |R|$, there is a vertex $y \in R \setminus \Gamma(S)$. All neighbours of y lie in $L \setminus S$, so

$$\deg(y) \leq |L| - |S| < |L| - \mathcal{D}(N) < \mathcal{D}(N),$$

contradicting the reverse minimum-degree bound. Hall's theorem supplies the matching. Restricting it to any $B \subseteq C_N^{18}$ proves the last claim. $\square$

The half-minimum-degree argument is not needed below the preceding endpoint. Splitting both base classes by parity makes the square witnesses 2 and 5 identically inactive, while a Hall defect itself supplies a short triple. This gives a systemic matching down to the one-pivot scale.

Put

$$L_0 = \{x \leq N : x \equiv 18 \pmod{50}\}, \quad R_0 = \{y \leq N : y \equiv 7 \pmod{50}\},$$

and

$$L_1 = \{x \leq N : x \equiv 43 \pmod{50}\}, \quad R_1 = \{y \leq N : y \equiv 32 \pmod{50}\}.$$

Thus $C_N^{18} = L_0 \sqcup L_1$, $C_N = R_0 \sqcup R_1$, and $|L_i| \leq |R_i|$. Join L_i to R_i when the product plus one is squarefree.

Lemma 5.12 (Zero-coincidence three-form block). *Let x_1, x_2, x_3 be distinct elements of one L_i with $\max x_j - \min x_j \leq 200$. The number of $y \in R_i$ for which each of $x_1y + 1, x_2y + 1, x_3y + 1$ has a prime-square divisor at most 353 is less than*

$$0.003707 \left(\frac{N}{50} + 1 \right) + 1628952. \tag{20}$$

Proof. Let $\mathcal{P} = \{p : 3 \leq p \leq 353, p \neq 5\}$ and put $z_p = p^{-2}$. There are 69 such primes. No residue classes belonging to two different forms can coincide at a prime in $\mathcal{P}$: coincidence would give $p^2 \mid x_j - x_k$, whereas $50 \mid x_j - x_k$, so a nonzero difference would have absolute value at least $50p^2 \geq 450$.

For a list $w = (w_p)_{p \in \mathcal{P}}$ write

$$L_3(w) = 1 - e_1(w) + e_2(w) - e_3(w), \quad U_2(w) = 1 - e_1(w) + e_2(w),$$

where the e_j are elementary symmetric functions. The degree-3 lower and degree-2 upper Bonferroni inequalities applied to the survivor events give the common-bad density at most

$$I_3 := 1 - 3L_3(z) + 3U_2(2z) - L_3(3z).$$

Exact rational evaluation gives

$$\sum_{p \in \mathcal{P}} \frac{1}{p^2} < 0.161841, \quad I_3 < 0.003707. \quad (21)$$

Every nonconstant CRT term has one terminal incomplete class. The total number of such terms in the displayed Bonferroni combination is

$$3 \sum_{j=1}^3 e_j(1, \dots, 1) + 3 \sum_{j=1}^2 e_j(2, \dots, 2) + \sum_{j=1}^3 e_j(3, \dots, 3) = 1628952.$$

This proves (20). □

We use Dusart's explicit estimate [4]

$$\pi(y) \leq \frac{y}{\log y - 1.1} \quad (y > 60184). \quad (22)$$

For $N \geq 2 \cdot 10^9$ define

$$P(N) = \frac{N}{19(\log(N/20) - 1.1)}, \quad \tau = 0.000407315,$$

and let

$$H(N) = \begin{cases} 1024, & 2 \cdot 10^9 \leq N < 2587877292, \\ 2048, & 2587877292 \leq N < 7.67769 \cdot 10^9. \end{cases}$$

Put

$$E_H(N) = \left(\frac{N}{50} + 1 \right) \tau + P(N) + 400.801H(N), \quad (23)$$

$$D_H(N) = \frac{N}{50} - 1 - 0.161841 \left(\frac{N}{50} + 1 \right) - 69 - E_H(N), \quad (24)$$

$$U_H(N) = 0.003707 \left(\frac{N}{50} + 1 \right) + 1628952 + 3E_H(N). \quad (25)$$

Lemma 5.13 (Uniform medium-prime payment in the parity blocks). *On $2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9$, every vertex of each graph L_i - R_i has degree at least $D_H(N)$, in both directions. Moreover, every triple in Lemma 5.12 has common non-neighbourhood in R_i of size at most $U_H(N)$, and*

$$D_H(N) > \frac{N}{100}, \quad U_H(N) < D_H(N). \quad (26)$$

Proof. For a pivot t in either block and a variable in the opposite block, 2 and 5 cannot be square witnesses. For primes through 353, the union bound and (21) give the first two subtractions in (24).

For the remaining primes take $Y = \lfloor N/19 \rfloor$. The density part is at most $(N/50 + 1)\tau$, since the rational prime-square tail calculation gives

$$\sum_{p>353} \frac{1}{p^2} < 0.000407315.$$

For $353 < p \leq Y$, pay one terminal class per prime. Dusart's estimate (22), together with $N/20 \leq Y \leq N/19$, gives

$$\pi(Y) \leq P(N).$$

For $p > Y$, write the variable as $r + 50n$. The transformed progression has modulus $50t$ and coprime initial term $rt + 1$. Removing one redundant factor 5 leaves a root modulus at most $10N$. The least modulus with more than 1024 unit square roots is

$$8P_9 = 25878772920,$$

and the least with more than 2048 is

$$8P_{10} = 802241960520.$$

This proves the displayed step bound $H(N)$. Repeating the elementary hyperbola split that gives (16), now with a parity-block progression of length at most $N/50 + 1$, bounds the large-prime terminal classes by

$$H(N) \left(\frac{2(N/50 + 2)}{Y} + \frac{N^2 + 50N + 1}{Y^2} \right) < 400.801H(N).$$

Thus the full tail is bounded by $E_H(N)$, proving the degree assertion. For a common non-neighbour of three pivots, either all witnesses are at most 353, when Lemma 5.12 applies, or at least one form uses a larger witness; the union bound pays $3E_H(N)$. This gives (25).

With $H = 1024$, direct substitution at $N = 2 \cdot 10^9$ gives

$$\frac{D_H(N)}{N} > 0.01351113, \quad \frac{U_H(N)}{N} < 0.01064464.$$

At the sole root jump, $N = 2587877292$ and $H = 2048$, it gives

$$\frac{D_H(N)}{N} > 0.01344370, \quad \frac{U_H(N)}{N} < 0.01066191.$$

On each constant- H interval, $D_H(N)/N$ increases and $U_H(N)/N$ decreases. The two endpoint checks prove (26). $\square$

Theorem 5.14 (Systemic pure-18 parity matching). *For every $N \geq 2 \cdot 10^9$, the squarefree-edge graph between C_N^{18} and C_N has a matching that saturates C_N^{18} . Consequently every outside set $B \subseteq C_N^{18}$ satisfies the original Hall inequality (2).*

Proof. The range $N \geq 7.67769 \cdot 10^9$ is Corollary 5.11, so consider the preceding interval. Fix one block $L_i - R_i$ and suppose $S \subseteq L_i$ is a Hall defect. Write $m = |S|$. If $m \leq D_H(N)$, the neighbourhood of one member of S already has size at least m , a contradiction. Hence

$$m > D_H(N) > \frac{N}{100}.$$

List S increasingly as $x_1 < \dots < x_m$. Since

$$\sum_{j=1}^{m-2} (x_{j+2} - x_j) \leq 2(N-1),$$

some three consecutive members have span less than 201, and hence at most 200. Lemma 5.13 bounds their common non-neighbourhood by $U_H(N) < D_H(N)$.

On the other hand let $T = R_i \setminus \Gamma(S)$. Every neighbour of a vertex of T lies in $L_i \setminus S$, so the reverse degree bound gives $m \leq |L_i| - D_H(N)$. The assumed Hall defect therefore implies

$$|T| > |R_i| - m \geq |R_i| - |L_i| + D_H(N) \geq D_H(N),$$

contradicting the common-non-neighbourhood bound. Hall's theorem supplies a matching in each parity block. Their disjoint union saturates $C_N^{18} = L_0 \sqcup L_1$. Restricting the matching to any $B \subseteq C_N^{18}$ proves the last statement. $\square$

The exact rational constants and both root-jump endpoint margins are reproduced by

`scripts/systematic_pure_matching_close.py`.

We next retain the quadratic-residue screen in the one-pivot progression itself. This is the ingredient needed to make the preceding matching robust against a genuinely mixed outside set.

Put

$$X_0(N) = \frac{501}{25} + \frac{501}{N}, \quad M_0 = 251002, \quad \eta_0 = 0.002267575.$$

For $0 \leq k \leq 8$, define s_k by

k	0	1	2	3	4	5	6	7	8
s_k	251002	251002	251002	125464	63250	32653	16510	8125	3679.

Proposition 5.15 (QR-correlated mixed degree). *For*

$$2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9,$$

every outside vertex has more than

$$\frac{N}{50} + 4535153 \tag{27}$$

squarefree neighbours in C_N . The same lower bound holds in the reverse progression from C_N into C_N^{18} . More precisely, the normalized excess in (27) is greater than 0.0022675765 at the lower endpoint and increases on the interval.

Proof. Take $D = \lfloor N/500 \rfloor$. Then $D \geq N/501$, and hence

$$\frac{X}{D} \leq X_0(N), \quad \frac{N^2 + 25N + 1}{D^2} < 251002 = M_0.$$

Let k be the number of non-5 odd prime divisors of the effective root modulus. Nine such divisors are impossible, because

$$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 = 20056049013 > N.$$

If the 2-adic exponent is ν , the root count is at most

$$H_{\nu,k} = c_\nu 2^{k+1}, \quad c_0 = c_1 = 1, \quad c_2 = 2, \quad c_\nu = 4 \ (\nu \geq 3).$$

For $k \geq 3$, product forcing supplies the following small subsets of the odd prime divisors:

k	cutoff	forced subset	$\max \#\{m \leq M_0 : m \text{ survives}\}$
3	2000	1	125464
4	1400	2	63250
5	1000	3	32653
6	360	4	16510
7	160	5	8125
8	60	6	3679.

Indeed, failure of the first two assertions would force respectively the products

$$2003 \cdot 2011 \cdot 2017 = 8124542561, \quad 3 \cdot 1409 \cdot 1423 \cdot 1427 = 8583434967,$$

and the remaining four excluded products are the ones displayed in the proof of Proposition 5.8; all exceed the upper endpoint. Every contributing complementary variable m is a nonzero quadratic residue modulo each selected prime divisor. Exact intersections of those residue masks give the last column. For $k \leq 2$ we simply retain all M_0 values, giving the first three entries $s_k = M_0$.

The same Möbius split used in Proposition 5.9 therefore gives, for an odd pivot, the lower bound

$$G_{0,k}(N) = \frac{\kappa_5(N-43)}{25} - \frac{N}{500} - X_0(N) - H_{0,k} \left(s_k + \frac{X_0(N)}{2} \left(4 + \frac{s_k}{3026} \right) \right) - \frac{501}{5N} - 1. \quad (28)$$

For an even pivot the Euler product omits the factor at 2, so the first term is instead $4\kappa_5(N-43)/75$; use the corresponding c_ν in $H_{\nu,k}$. These bounds discard the modulus-5 and 2-adic residue screens and every further Euler gain.

The term $s_k/3026$ cannot be discarded. The reciprocal sum over the permitted transformed parameters through 3025 is at most 4, while each remaining survivor contributes at most $1/3026$. Therefore

$$\sum_{m \text{ surviving}} \frac{1}{m} \leq 4 + \frac{s_k}{3026}.$$

The elementary transformed-parameter count then gives the root term in (28).

There are only $4 \cdot 9$ displayed valuation/root strata. Direct exact substitution at $N = 2 \cdot 10^9$ shows that the minimum is the odd $k = 6$ stratum:

$$G_{0,6}(N) - \frac{N}{50} > 4535153.1398.$$

The next smallest odd strata retain more than $4.55 \cdot 10^6$, and every even stratum retains more than $1.50 \cdot 10^7$. In each fixed stratum the affine main term increases, while $X_0(N)$ and all normalized constant payments decrease. Feasible prime supports only disappear as N decreases. Thus the lower endpoint controls the full interval. The reverse 18 (mod 25) progression has the identical root and residue-screen calculation. This proves the proposition. $\square$

The support forcing, all exact QR-mask intersections, and all 36 endpoint comparisons are reproduced by

`scripts/systematic_mixed_degree_close.py.`

The reciprocal arithmetic is checked by

`scripts/verify_four_range_paper_arithmetic.py.`

Corollary 5.16 (A systemic mixed residual). *Let B be a Hall defect in the interval of Proposition 5.15, and put $B^* = B \setminus C_N^{18}$. Then*

$$|B^*| > 0.00453515N. \quad (29)$$

Consequently B^ , and hence $A^* = (B \cup T_N(B)) \setminus (C_N \cup C_N^{18})$, contains a mod-4 class with three members of span less than 1764.*

Proof. The pure case is already excluded by Theorem 5.14, so choose $x \in B^*$. Compatibility and the reverse degree bound give

$$|B \cap C_N^{18}| < |C_N^{18}| - \frac{N}{50} - 4535153,$$

whereas Hall defect and the forward degree give $|B| > N/50 + 4535153$. Since $|C_N^{18}| \leq N/25 + 1$,

$$|B^*| > 9070305 > 0.00453515N$$

at the lower endpoint; the normalized lower bound increases thereafter by the preceding proof.

Let S be the largest mod-4 class in B^* . Then $|S| > 0.0011337875N$. If $x_1 < \dots < x_m$ are its members, then

$$\sum_{j=1}^{m-2} (x_{j+2} - x_j) \leq 2(N-1).$$

At the lower endpoint this forces a three-term span below 1764, and the quotient decreases thereafter. Hence an integral span is at most 1763. $\square$

Lemma 5.17 (Three-term prime count and integrated square tail). *For $y \geq 355991$,*

$$\pi(y) \leq \frac{y}{\log y} \left(1 + \frac{1}{\log y} + \frac{2.51}{\log^2 y} \right), \quad \sum_{p>131} \frac{1}{p^2} < 0.001257623.$$

Consequently, if $L = \log(N/(k+1))$, the intermediate-prime payment in a two-pivot tail is at most

$$\frac{4}{k} \left(\frac{1}{L} + \frac{1}{L^2} + \frac{2.51}{L^3} \right).$$

Proof. The prime-count estimate is Theorem 6.9 of [4]. Exact rational summation gives

$$\sum_{131 < p \leq 10^5} \frac{1}{p^2} < 0.0012554421.$$

Partial summation and $\pi(t) < 1.25506t/\log t$ give

$$\sum_{p > 10^5} \frac{1}{p^2} \leq 2(1.25506) \int_{10^5}^{\infty} \frac{dt}{t^2 \log t} < 0.0000021803.$$

This proves the square-tail constant. Since $N/(k+1) \leq \lfloor N/k \rfloor \leq N/k$, monotonicity of the prime-count majorant gives the normalized payment. $\square$

Lemma 5.18 (Sharp negative-Pell gap). *Let $d \geq 2$ be nonsquare. The number of positive solutions of*

$$x^2 - dy^2 = -1, \quad x \leq N,$$

is at most

$$\left\lfloor 1 + \log_{3+2\sqrt{2}} N \right\rfloor. \quad (30)$$

In particular this number is at most 13 when $2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9$.

Proof. For a solution put $\alpha = x + y\sqrt{d}$. If $\alpha_2 > \alpha_1$ are two solutions, then $\alpha_2/\alpha_1 = u + v\sqrt{d}$ is an integral totally positive unit of norm 1. Here $u \geq 2$, while $u = 2$ would force $d = 3$ and then $x^2 \equiv -1 \pmod{3}$, which is impossible. Hence consecutive values of α have ratio at least $3 + 2\sqrt{2}$. Also

$$1 + \sqrt{2} \leq \alpha = x + \sqrt{x^2 + 1} \leq (1 + \sqrt{2})x,$$

so the ratio of the largest to the smallest permitted value is at most N . This proves (30). Finally

$$(3 + 2\sqrt{2})^{12} < 2 \cdot 10^9 < 7.67769 \cdot 10^9 < (3 + 2\sqrt{2})^{13}.$$

$\square$

We now give the terminal three-pivot payment. Set

$$\tau_{131} = 0.001257623, \quad P_- = 13, \quad H_{\Sigma} = 384.$$

The first constant is Lemma 5.17, the second is Lemma 5.18, and the last follows from the QR hierarchy: for $Y = \lfloor N/77 \rfloor$, each pivot's high-root tail is no larger than the root-128 unscreened tail. Indeed $m \leq 5929$, and the four higher-root mask maxima are 986, 592, 315, 146; in every case

$$H \left(\#m + \frac{N/25 + 2}{2Y} \left(4 + \frac{\#m}{3026} \right) \right) \leq 128 \left(5929 + \frac{N/25 + 2}{2Y} \left(4 + \frac{5929}{3026} \right) \right).$$

The four mask maxima remain valid; their comparison is checked by `scripts/verify_four_range_paper_arithmetic.py`.

The exact restricted Möbius support in the aggregate diagonal estimate gives, uniformly on the present interval,

$$\Delta_{25}^{\text{mix}} < 0.002980825. \quad (31)$$

If three pivots lie in one mod-4 class which is globally concentrated in one mod-9 class, the allowed union of residue classes has density $161/225$; its prefix discrepancy modulo 900 is less than 5, and

$$\Delta_{900}^{\text{mix}} < 0.003766884. \quad (32)$$

The maximum in (31) occurs at

$$N = 2 \cdot 10^9, \quad R = 3809763, \quad \sum_{s \leq R} 2^{\omega(s)} = 787461$$

on the permitted squarefree semigroup. The factor 13 is used in both Pell tails.

For primes through 131, apply degree-3, degree-2, and degree-3 Bonferroni bounds to the one-, two-, and three-form survivors. A prime common to all three forms has multiplicity 1; a pair coincidence has multiplicity 1 on that pair and 2 in the three-form union. Concentrating all allowed pair coincidences on one edge maximizes the sum of the three quadratic pair bounds. Direct expansion gives the following terminal table. The column θ multiplies C_{quad} ; “charge” is an explicitly discarded part of A^* .

branch	θ	common primes	charge	(I, ϵ)	$ A /N$
even generic	23/25	7, 11, 13, 17	0	(0.041080150, 45066)	< 0.038927972
two odd classes	23/50	7, 11, 13, 17	η_0	(0.115095250, 49865)	< 0.034542160
two odd, common 3	23/50	3, 7, 11, 13, 17	η_0	(0.160187242, 48180)	< 0.038147835
one odd class	23/100	2, 7, 11, 13, 17	0	(0.290282102, 48180)	< 0.039998152
one odd, common 2, 3	23/900	2, 3	0.0012	(0.344765994, 53410)	< 0.039971251
global mod 9	23/225	3, 7, 11, 13, 17	0	(0.145156038, 43495)	< 0.024889106
even concentrated	161/225	3	0	(0.113917972, 50119)	< 0.039955268
mixed parity	23/100	7, 11, 13, 17	η_0	(0.261748734, 49285)	< 0.039984162

Here every row includes

$$\frac{6\tau_{131}}{25} < 0.000301830$$

and the common intermediate-prime/root payment

$$\begin{aligned} & \frac{6}{77} \left(\frac{1}{L} + \frac{1}{L^2} + \frac{2.51}{L^3} \right) + \frac{2H_\Sigma}{N} \left(5929 + \frac{156}{25} + \frac{312}{N} + \left(\frac{39}{25} + \frac{78}{N} \right) \frac{5929}{3026} \right) \\ & < 0.007155053, \quad L = \log(N/78). \end{aligned}$$

Thus the displayed values are strict upper bounds at $N = 2 \cdot 10^9$; every analytic term decreases thereafter. The exact rational finite polynomials and restricted diagonal jump scan are reproduced by `scripts/systematic_mixed_full_close.py`; all branch totals are checked by `scripts/verify_four_range_paper_arithmetic.py`.

Theorem 5.19 (Systemic mixed three-pivot close). *The Hall inequality (2), and hence the required bound for Erdős Problem 848, holds for*

$$2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9.$$

Consequently it holds for every $N \geq 2 \cdot 10^9$.

Proof. Suppose that B is a Hall defect and put $A = B \cup T_N(B)$. Then A is compatible, $|A| > |C_N|$, and A^* contains B^* from Corollary 5.16. Let e be the number of even members of A^* and split at $e = \eta_0 N$.

First suppose $e > \eta_0 N$. One even mod-4 class contains more than $0.0011337875N$ members. If that class is not contained in one residue modulo 9, choose three of its members which are nonconstant modulo 9. The prime 2 is inactive and 3 is not common. Since a common-prime product Q satisfies $4Q^2 < N$, at most four further common primes occur; replacing them by 7, 11, 13, 17 only enlarges the finite survivor bound. This is the first row of the table.

If the even class is contained in one residue modulo 9, choose three consecutive members. Their span is at most 1763, so 3 is the only possible common odd prime because $4(3 \cdot 7)^2 = 1764$. The other three mod-4 classes are unrestricted, while the selected class is restricted modulo 9; this is precisely the 161/225 residue union in (32). The seventh row closes this case.

Now suppose $e \leq \eta_0 N$. Charge the even part. If the odd part meets both odd classes modulo 4, choose a triple nonconstant modulo 4; if the odd part is also nonconstant modulo 9, the same triple may be chosen nonconstant for both projections. These are respectively the second and, when all odd members share one residue modulo 9, the third rows. The diagonal set is restricted to odd non-base values, giving $\theta = 23/50$.

It remains that all odd members of A^* occupy one mod-4 class. If $e = 0$ and this class is nonconstant modulo 9, the prime 2 is common but 3 is not; the fourth row applies. If the class is constant modulo 9, three consecutive odd pivots have common primes only 2 and 3. When $e \leq 0.0012N$, charge the even part and use the fifth row. When instead $e > 0.0012N$ and all of A^* is constant modulo 9, select three even pivots; the prime 2 is inactive, and the sixth row allows the common prime 3 and four further common primes.

Finally suppose $0 < e \leq \eta_0 N$ and A^* is not constant modulo 9. Choose two odd pivots and one even pivot so that the three residues modulo 9 are nonconstant. The 2-square event is active on at most two forms and is not common. The asymmetric finite polynomial in the last row, together with the charged even part, closes this final case.

These alternatives are exhaustive. Every row gives

$$\frac{|A|}{N} < \frac{1}{25} - \frac{7}{25N} \leq \frac{|C_N|}{N},$$

contradicting $|A| > |C_N|$. This proves the stated interval. The range $N \geq 7.67769 \cdot 10^9$ follows from the independently certified high-tail Theorem 6.31, completing the last assertion. $\square$

5.1 Actual-support descent to two hundred million

For $N_0 \leq N < N_1$ we retain the ordered prime support. Put

$$N_0 = 2 \cdot 10^8, \quad N_1 = 2 \cdot 10^9, \quad D = \lfloor N/300 \rfloor, \quad M = 90001.$$

If a pivot has k distinct non-5 odd prime divisors, retain its smallest $k - 1$ of them and require that a larger k -th prime still fit under N_1 . Exact intersections of their nonzero quadratic-residue masks on $m \leq M$ give

k	0	1	2	3	4	5	6	7	8
s_k	90001	90001	45180	22924	11414	5520	2513	1039	353.

The maximizing supports for $k = 2, \dots, 8$ are respectively

$$(38639), (971, 1429), (137, 193, 199), (43, 61, 73, 89), \\ (23, 29, 31, 41, 43), (11, 13, 17, 19, 23, 41), (3, 7, 13, 17, 19, 23, 29).$$

Proposition 5.20 (Uniform actual-support degree). *For $N_0 \leq N < N_1$, every outside vertex has more than*

$$\frac{N}{50} + 0.0001574N$$

squarefree neighbours in C_N , and the identical reverse bound holds from C_N into C_N^{18} . Consequently every mixed Hall defect satisfies

$$|B \setminus C_N^{18}| > 0.0003148N. \quad (33)$$

Proof. Since $D \geq (N - 299)/300$, throughout the interval

$$\frac{N^2 + 25N + 1}{D^2} < 90001.$$

For a valuation factor $c_\nu \in \{1, 2, 4\}$ and k odd prime divisors, put $H = c_\nu 2^{k+1}$. The same Möbius split as in Proposition 5.15, now with the actual-support survivor s_k , gives

$$\begin{aligned} \deg(b) - \frac{N}{50} &\geq \left(\frac{4}{3}\right)^{[2b]} \frac{\kappa_5(N-43)}{25} - \frac{N}{300} - \frac{N}{50} - 1 \\ &\quad - \frac{N/25 + 1}{(N-299)/300} - \frac{1}{5(N-299)/300} \\ &\quad - H \left(s_k + \frac{X_D}{2} \left(4 + \frac{s_k}{3026} \right) \right), \quad X_D = \frac{N/25 + 1}{(N-299)/300}. \end{aligned}$$

Every fixed row divided by N is increasing: its leading coefficient is positive, while each remaining positive payment divided by N decreases. It therefore suffices to use $N = N_0$ and the support superset at N_1 . Among all $4 \cdot 9$ rows the minimum is the odd $k = 3$, $H = 16$ row, with survivor 22924 and witness (971, 1429); its margin is

$$31482.2990360 \dots > 0.0001574N_0.$$

The reverse progression has the same support and root calculation. The argument of Corollary 5.16 then gives $|B \setminus C_N^{18}| > 2(0.0001574N) - 1$, proving (33). The exact support enumeration is reproduced by `scripts/systematic_mixed_interval_support.py`; the degree rows are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

The transformed congruence contains one further uniform condition which was not needed for Proposition 5.20. In the notation of the progression split, $d_1 = 25b/g$ with $g \in \{1, 5\}$, so $5 \mid d_1$; moreover c_1 is a unit. Hence

$$d^2 m \equiv c_1 \pmod{d_1}$$

forces m into one fixed nonzero square coset modulo 5. Taking the worse of the two possible cosets and intersecting it with the same actual-support masks gives

k	0	1	2	3	4	5	6	7	8
$s_k^{(5)}$	36001	36001	18234	9314	4680	2290	1055	471	175.

The maximizing retained supports for $k = 2, \dots, 8$ are respectively

$$\begin{aligned} &(44563), (1087, 1117), (101, 241, 263), (53, 59, 71, 83), \\ &(19, 29, 37, 41, 43), (11, 13, 17, 19, 29, 31), (3, 7, 11, 13, 17, 23, 37). \end{aligned}$$

Corollary 5.21 (Mod-5-coset actual-support degree). *On $N_0 \leq N < N_1$, every forward and reverse outside degree is greater than*

$$\frac{N}{50} + 0.0012427N.$$

Consequently every mixed Hall defect in this interval satisfies

$$|B \setminus C_N^{18}| > 0.0024854N - 1 > 0.002485N.$$

Proof. Repeat Proposition 5.20 with $s_k^{(5)}$. The unique controlling row is now the odd $k = 4$ row, with $H = 32$, survivor 4680, and retained support (101, 241, 263). At $N = N_0$ its margin above $N/50$ is

$$248552.6176834 \dots > 0.0012427N_0.$$

The same normalized monotonicity proves the degree bound on the full interval. Applying the forward–reverse defect argument gives the displayed residual. Both cosets and every feasible support prefix at the upper endpoint are enumerated by `scripts/systematic_mod5_actual_support.py`; the degree rows are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

The same support constraint is retained in the large-square tail. On the following four blocks use $Y = \lfloor N/k_0 \rfloor$. The integer box is $m \leq k_0^2$: indeed $(k_0^2 + 1)(N/k_0 - 1)^2 > N^2 + 25N + 1$ at the left endpoint, and the difference increases. Exact support intersections give the displayed worst root row and the resulting three-pivot prime/root ceiling; the intermediate-prime term uses the three-term Dusart bound and $Y \geq N/(k_0 + 1)$.

$[L, U)$	k_0	H	s	support	three-pivot ceiling
$[2, 3] \cdot 10^8$	75	128	805	(97, 113, 151)	0.008955907
$[3, 5] \cdot 10^8$	85	128	1016	(97, 113, 151)	0.007668494
$[5, 10] \cdot 10^8$	100	128	1391	(73, 199, 257)	0.006336343
$[10, 20] \cdot 10^8$	125	128	2124	(47, 233, 401)	0.004881897.

All support maxima are reproduced by `scripts/systematic_mod5_actual_support.py`. The monotone analytic payments are checked by `scripts/verify_four_range_paper_arithmetic.py`.

Theorem 5.22 (Actual-support pure-18 parity matching). *For every $N \geq 2 \cdot 10^8$, the squarefree-edge graph between C_N^{18} and C_N has a matching saturating C_N^{18} . Consequently every $B \subseteq C_N^{18}$ satisfies the original Hall inequality.*

Proof. It remains only to treat $N_0 \leq N < N_1$. Work in the parity blocks L_i – R_i used in Theorem 5.14. The three-pivot ceiling above is six times a one-form intermediate-prime plus root payment. A parity-block progression has length $N/50 + 1$, rather than $N/25 + 1$, so one sixth of that ceiling is a valid one-form bound. Add the square-density tail $(N/50 + 1)0.000407315$ and retain the exact finite constants of Lemma 5.12. At the four left endpoints, the degree lower bound D and common-non-neighbour upper bound U satisfy

N	D/N	U/N	$D - U$
$2 \cdot 10^8$	0.0152620317	0.0127212769	> 508150
$3 \cdot 10^8$	0.0154767174	0.0093626504	> 1834220
$5 \cdot 10^8$	0.0156988362	0.0065246389	> 4587098
10^9	0.0159413140	0.0041684639	> 11772850 .

On each block D/N increases and U/N decreases. In particular $D > N/100$ and $U < D$. A Hall defect in either parity block therefore contains three consecutive pivots of span at most 200; the reverse degree argument forces its common non-neighbourhood to exceed D , contradicting $U < D$, exactly as in Theorem 5.14. The four endpoint calculations are reproduced by `scripts/verify_four_range_paper_arithmetic.py`. The already proved range $N \geq N_1$ completes the assertion. $\square$

We finally replace the absolute small-square rounding loss by an exact finite prefix. Let $R = 2 \cdot 10^7$. For a union W of non-base residue classes, put

$$Q_W(U) = - \sum_{\substack{2 \leq s \leq R \\ p|s \Rightarrow p \equiv 1 \pmod{4}}} \mu(s) \# \{x \leq U : x \in W, s^2 \mid x^2 + 1\} \\ + \frac{2\theta_W U (\log R + 2)}{R} + 13 \frac{U^2 + 1}{R^2},$$

where θ_W is the density of W . Möbius inversion, the harmonic tail, and the sharp negative-Pell orbit bound show that $Q_W(U)$ is an upper bound for the diagonal non-squarefree count through U .

Partition $[N_0, N_1)$ into 463 adjacent blocks $[L, U]$, each of relative width less than 0.005, with additional breaks at $3 \cdot 10^8, 5 \cdot 10^8, 10^9$. Since the actual diagonal count is monotone, $Q_W(U)/L$ bounds its normalized value throughout the block. The finite computation enumerates 1237599 squarefree support nodes, 4134027 roots, and 51357374 progression events. The eighteen concentrated sets are evaluated simultaneously from

$$U_{25} - E_e + E_{e,c} \quad (e = 0, 2; c \pmod{9}).$$

The controlling block is $[200000000, 200999999]$, where the worst concentrated diagonal is

$$\frac{Q_W(U)}{L} < 0.019673927941605. \quad (34)$$

All integer events and block maxima are reproduced by `scripts/systematic_concentrated_diagonal_blocks.cpp`.

Theorem 5.23 (Actual-support mixed close). *The Hall inequality (2), and hence the exact bound for Erdős Problem 848, holds for every*

$$N \geq 2 \cdot 10^8.$$

Proof. The range $N \geq N_1$ is Theorem 5.19 and the subsequent descent, so let $N_0 \leq N < N_1$. Suppose B is a Hall defect and put $A = B \cup T_N(B)$. Theorem 5.22 excludes $B \subseteq C_N^{18}$. Corollary 5.21 therefore gives

$$|A^*| \geq |B \setminus C_N^{18}| > 0.002485N.$$

Let e be the number of even members of A^* and put $\eta = 0.00016$.

If $e > \eta N$, one even mod-4 class has density greater than $\eta/2$. A closest pair in it has gap less than $2/\eta = 12500$. If the class is nonconstant modulo 9, add a third pivot which makes the triple nonconstant modulo 9; the prime 2 is inactive and 3 is not common. If it is constant modulo 9, the same short pair shows that the only possible common odd primes are 3, 7, since

$$4(3 \cdot 7 \cdot 11)^2 = 213444 > 12500.$$

These are the even-generic and even-concentrated rows below.

Now suppose $e \leq \eta N$ and charge all even members. More than $0.002325N$ odd members remain. If both odd mod-4 classes occur, choose a triple nonconstant modulo 4 and split according as its odd support is nonconstant or constant modulo 9. If only one odd mod-4 class occurs, a closest pair has gap less than 6169.4. In the nonconstant-mod-9 case add a third pivot of another mod-9 residue: 2 is common, 3 is not, and at most one further odd prime is common because $4(7 \cdot 11)^2 = 23716 > 6169.4$. In the constant case the common list is at most 2, 3, 7, since adjoining 11 would again force a gap at least $4(3 \cdot 7 \cdot 11)^2$.

Use cutoff 131, degree-3/2/3 Bonferroni bounds, the integrated tail $6\tau_{131}/25$, the appropriate exact diagonal prefix, and the root ceiling of the containing product block. Exhausting all 463 blocks gives

branch	allowed common primes	$\max A /N$
even generic	7, 11, 13, 17	0.038287487920
even concentrated	3, 7	0.039870783152
two odd classes	7, 11, 13, 17	0.031773851295
two odd, common 3	3, 7, 11, 13, 17	0.035364360630
one odd class	2, 7	0.038964611892
one odd, common 2, 3	2, 3, 7	0.039088651986.

Every maximum occurs in the first block. Its exact target is $1/25 - 7/(25N) > 0.0399999986$, so all six exhaustive alternatives contradict $|A| > |C_N|$. This proves the new interval and hence the theorem. $\square$

Corollary 5.24 (A mixed-base strip). *Let $N \geq 7.67769 \cdot 10^9$ and let B be a compatible outside set. Put*

$$B^* = B \setminus C_N^{18}, \quad \mathcal{R}(N) = 2\mathcal{D}(N) - |C_N^{18}|.$$

If $|B^| \leq \mathcal{R}(N)$, then B satisfies the Hall inequality. Moreover*

$$\mathcal{R}(N) > 0.000000036N \quad (2.2827 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and, more sharply,

$$\mathcal{R}(N) > 0.0010038N \quad (7.67769 \cdot 10^9 \leq N < 1.15 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.0000268N \quad (1.15 \cdot 10^{10} \leq N < 1.2643 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.000000228N \quad (1.2643 \cdot 10^{10} \leq N < 1.279 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.00000079N \quad (1.279 \cdot 10^{10} \leq N < 1.29 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.00000104N \quad (1.29 \cdot 10^{10} \leq N < 1.382 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.00000236N \quad (1.382 \cdot 10^{10} \leq N < 2.2827 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.000127N \quad (2.3669 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001162N \quad (3.22715 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001187N \quad (3.2532 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001414N \quad (3.4984 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001424N \quad (3.51 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.00032N \quad (1.001 \cdot 10^{11} \leq N < N_{25}),$$

and

$$\mathcal{R}(N) > 0.000724N \quad (N \geq 5.5 \cdot 10^{11}).$$

Thus every Hall defect in the new bottom interval must contain more than $0.001627N$ genuinely non-base outside vertices. The subsequent displayed bounds give the strip separately on each later degree piece.

Proof. If $B^* = \emptyset$, use Corollary 5.11. Otherwise fix $x \in B^*$. Repeat the relevant one-pivot degree proof with the variable in the progression $18 \pmod{25}$. Since $x \not\equiv 18 \pmod{25}$, the gcd $(18x + 1, 25x)$ is 1 or 5, so the identical calculation gives at least $\mathcal{D}(N)$ values $y \in C_N^{18}$ for which $xy + 1$ is squarefree. Compatibility forces all of them out of B . Consequently

$$|B \cap C_N^{18}| \leq |C_N^{18}| - \mathcal{D}(N).$$

Under the stated hypothesis,

$$|B| \leq |C_N^{18}| - \mathcal{D}(N) + |B^*| \leq \mathcal{D}(N).$$

The corresponding one-pivot degree bound now proves the Hall inequality.

Finally $|C_N^{18}| \leq N/25 + 1$. Direct substitution in the displayed formula for $\mathcal{M}_{\text{QR2}}$ at $N_0 = 7.67769 \cdot 10^9$ gives

$$\frac{2\mathcal{M}_{\text{QR2}}(N_0)}{N_0} - \frac{1}{25} - \frac{1}{N_0} > 0.0010038.$$

Direct substitution in $\mathcal{M}_{\text{QR}}$ at $1.15 \cdot 10^{10}$ gives

$$\frac{2\mathcal{M}_{\text{QR}}(1.15 \cdot 10^{10})}{1.15 \cdot 10^{10}} - \frac{1}{25} - \frac{1}{1.15 \cdot 10^{10}} > 0.0000268.$$

Direct substitution in $\mathcal{M}_{1/32}$ at $1.2643 \cdot 10^{10}$ gives

$$\frac{2\mathcal{M}_{1/32}(1.2643 \cdot 10^{10})}{1.2643 \cdot 10^{10}} - \frac{1}{25} - \frac{1}{1.2643 \cdot 10^{10}} > 0.000000228.$$

Direct substitution in $\mathcal{M}_{1/16}$ at $1.279 \cdot 10^{10}$ gives the stronger 0.00000079 bound. Direct substitution in $\mathcal{M}_{1/8}$ at $1.29 \cdot 10^{10}$ gives the stronger 0.00000104 bound. Direct substitution in $\mathcal{M}_{1/4}$ at $1.382 \cdot 10^{10}$ gives the stronger 0.00000236 bound. Direct substitution in $\mathcal{M}_{1/2}$ at $2.2827 \cdot 10^{10}$ gives the weaker transition bound 0.000000036, while substitution at $2.3669 \cdot 10^{10}$ gives the sharper 0.000127 bound. Direct substitutions at $3.22715 \cdot 10^{10}$, $3.2532 \cdot 10^{10}$, $3.4984 \cdot 10^{10}$, and $3.51 \cdot 10^{10}$ give

the sharper 0.001162, 0.001187, 0.001414, and 0.001424 bounds. The quotient increases on each of the displayed degree pieces. Direct checks from $\mathcal{M}_{1/2}$ to $\mathcal{M}_1$ preserve both displayed bounds on their stated intervals. The later switches to $\mathcal{M}$, across its root-count jump, and from $\mathcal{M}$ to $\mathcal{L}$ keep the stated weaker bounds. Direct evaluation at $1.001 \cdot 10^{11}$ gives more than $0.00032N$, and at $5.5 \cdot 10^{11}$ gives more than $0.0012N$, and the same switch checks prove the $0.000724N$ bound from that point onward. $\square$

6 The valid two-pivot CRT decomposition

The large-prime tail is nonempty in general, but it can be placed into exact CRT classes without changing the Hall target.

Lemma 6.1 (Two-pivot square classes). *Let $b, c \not\equiv 7 \pmod{25}$ be distinct, and let $a \equiv 7 \pmod{25}$. Suppose both $ab + 1$ and $ac + 1$ are non-squarefree. Choose primes p, q with*

$$p^2 \mid ab + 1, \quad q^2 \mid ac + 1.$$

Then $p, q \neq 5$, and exactly one of the following alternatives applies:

1. $p = q$ and $p^2 \mid b - c$;
2. $p \neq q$ and a lies in one uniquely determined residue class modulo $25p^2q^2$.

Proof. If $25 \mid ab + 1$, then $7b + 1 \equiv 0 \pmod{25}$. Since $7^{-1} \equiv 18 \pmod{25}$, this forces $b \equiv 7 \pmod{25}$, contrary to the hypothesis. Hence $p \neq 5$, and similarly $q \neq 5$.

If $p = q$, subtracting the two divisibilities gives $p^2 \mid a(b - c)$. But $p \nmid a$, since $p \mid a$ would imply $ab + 1 \equiv 1 \pmod{p}$. Therefore $p^2 \mid b - c$.

If $p \neq q$, the three congruences

$$a \equiv 7 \pmod{25}, \quad a \equiv -b^{-1} \pmod{p^2}, \quad a \equiv -c^{-1} \pmod{q^2}$$

have pairwise coprime moduli. The Chinese remainder theorem gives one class modulo $25p^2q^2$. $\square$

For a compatible pair $B = \{b, c\}$ define the same-prime and distinct-prime parts of its common non-neighbour set by

$$\begin{aligned} S_N(b, c) &= \{a \in T_N(B) : \text{some prime } p \text{ has } p^2 \mid ab + 1 \text{ and } p^2 \mid ac + 1\}, \\ D_N(b, c) &= T_N(B) \setminus S_N(b, c). \end{aligned}$$

This is an exact disjoint decomposition, not a new cut. The same-prime debit can be bounded independently of what occurs in $D_N(b, c)$.

Proposition 6.2 (Two-point same-prime debit). *Let $N \geq 107$ and let $B = \{b, c\}$ be a compatible outside set. Then*

$$2 + |S_N(b, c)| \leq |C_N|.$$

For $N \geq 500$ the stronger explicit estimate

$$|C_N| - 2 - |S_N(b, c)| > \frac{0.549N}{25} - \log_4(N - 1) - \frac{57}{25} \quad (35)$$

holds. Consequently the two-point Hall inequality follows whenever the distinct-prime part fits inside the right-hand side of (35).

Proof. By Lemma 6.1, every relevant prime satisfies $p \neq 5$ and $p^2 \mid |b - c|$. For a fixed p , the simultaneous conditions $a \equiv 7 \pmod{25}$ and $p^2 \mid ab + 1$ determine one residue class modulo $25p^2$. Hence

$$|S_N(b, c)| \leq \sum_{\substack{p^2 \mid |b-c| \\ p \neq 5}} \left(\frac{N}{25p^2} + 1 \right). \quad (36)$$

First suppose $107 \leq N \leq 499$. Since $(2 \cdot 3 \cdot 7)^2 = 1764 > N - 1$, at most two distinct primes occur in the sum, and the largest two class bounds are those for $p = 2, 3$. For $107 \leq N \leq 167$ the exact class counts are at most 2 and 1, so $2 + |S_N(b, c)| \leq 5 \leq |C_N|$. For $168 \leq N \leq 499$,

$$2 + |S_N(b, c)| \leq 4 + \frac{N}{100} + \frac{N}{225} = 4 + \frac{13N}{900} \leq \frac{N}{25} - \frac{7}{25} \leq |C_N|.$$

The middle inequality holds already at $N = 168$ and its right-minus-left side increases with N .

Now let $N \geq 500$, and let k be the number of primes in (36). Their squared product divides $|b - c| \leq N - 1$, so $4^k \leq N - 1$ and $k \leq \log_4(N - 1)$. Also, every relevant prime is different from 5. Removing 5 and the displayed composite integers from the sum over all integers gives the elementary numerical bound

$$\sum_{p \neq 5} \frac{1}{p^2} < \frac{\pi^2}{6} - 1 - \sum_{m \in \{4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}} \frac{1}{m^2} < 0.451.$$

It follows that

$$2 + |S_N(b, c)| < 2 + \frac{0.451N}{25} + \log_4(N - 1) \leq \frac{N}{25} - \frac{7}{25} \leq |C_N|.$$

At $N = 500$ the middle inequality has positive margin, and its margin is increasing because

$$\frac{0.549}{25} - \frac{1}{(N - 1) \log 4} > 0 \quad (N \geq 500).$$

This closes every case. Subtracting the last strict upper bound from $|C_N| \geq N/25 - 7/25$ gives (35). $\square$

Thus the same-prime contribution has a global debit bound, while every fixed finite distinct-prime skeleton is impossible. What remains is to fit the genuinely unbounded $D_N(b, c)$ contribution into the residual debit (35).

The extreme one-form tail already has a coefficient-uniform square-root bound. This pays the branch in which only one witness is very large, without requiring a second large witness.

Lemma 6.3 (Uniform one-form large-square tail). *For $1 \leq t \leq N$ and $Y \geq 1$, put*

$$\mathcal{H}_t(Y) = \{a \leq N : \text{some prime } p > Y \text{ satisfies } p^2 \mid at + 1\}$$

and

$$K_t = \left\lfloor \frac{tN + 1}{Y^2} \right\rfloor.$$

Then

$$|\mathcal{H}_t(Y)| \leq 2\sqrt{N + 1} \sum_{s=1}^{K_t} \frac{1}{\sqrt{s}} + 2K_t\sqrt{t} \leq 4\sqrt{K_t(N + 1)} + 2K_t\sqrt{t}. \quad (37)$$

The same bounds hold after restricting a to any progression.

Proof. First note the elementary root bound

$$\#\{x \pmod{m} : x^2 \equiv u \pmod{m}\} \leq 2\sqrt{m} \quad ((u, m) = 1). \quad (38)$$

Indeed, an odd prime power contributes at most two roots, while a power of 2 contributes 1, 2, or at most 4 roots according as its exponent is 1, 2, or at least 3. If the odd part of m has r distinct prime divisors, then

$$\frac{2^r}{\sqrt{\text{rad}(m_{\text{odd}})}} \leq \max \left\{ 1, \frac{2}{\sqrt{3}} \right\};$$

after the possible prime 3, every new odd prime contributes a factor at most $2/\sqrt{5} < 1$. Combining this with the three stated 2-adic cases gives (38). The case $m = 1$ is immediate.

For every counted a , choose a witnessing prime and write

$$p^2 s = at + 1.$$

Then $1 \leq s \leq K_t$, $(s, t) = 1$, and

$$p^2 \equiv s^{-1} \pmod{t}, \quad p \leq \sqrt{\frac{tN+1}{s}}.$$

For fixed s , (38) bounds the number of possible integers p —and hence the number of possible a —by

$$2\sqrt{t} \left(\frac{1}{t} \sqrt{\frac{tN+1}{s}} + 1 \right) \leq 2\sqrt{\frac{N+1}{s}} + 2\sqrt{t}.$$

Summing over s gives the first bound. The second follows from $\sum_{s \leq K} s^{-1/2} \leq 2\sqrt{K}$. Forgetting primality and progression conditions only enlarges the count. $\square$

Corollary 6.4 (The very-large mixed tail is paid). *Let $N \geq 41$ and $Y = \lfloor N/3 \rfloor$. For every $t \leq N$,*

$$|\mathcal{H}_t(Y)| < 27.42\sqrt{N+1}.$$

Consequently, for any three pivots $t_1, t_2, t_3 \leq N$, the number of $a \leq N$ for which at least one of the three forms $at_i + 1$ has a prime-square witness exceeding Y is less than

$$82.26\sqrt{N+1}. \quad (39)$$

This includes, in particular, the case in which exactly one witness exceeds Y .

Proof. The inequality $N^2 + 1 < 10\lfloor N/3 \rfloor^2$ holds for $N \geq 41$, so $K_t \leq 9$. Since

$$\sum_{s=1}^9 s^{-1/2} < 4.705,$$

the first bound in (37) is less than $9.41\sqrt{N+1} + 18\sqrt{N} < 27.42\sqrt{N+1}$. The union bound over the three forms gives (39). $\square$

The root count and tail arithmetic are reproduced by `scripts/uniform_large_square_tail_scan.py`. On its displayed finite box it checks 76116 unit-congruence cases and 173880 tail cases, with no root, tail, or $K_t \leq 9$ failure.

The large-square part of that obligation admits a coefficient-uniform Pell reduction. The point is to retain the determinant $|b - c|$ instead of multiplying one coefficient into the norm, which would introduce a spurious root count depending on the growing pivots.

Lemma 6.5 (Determinant-uniform Pell orbits). *Let A, B, d be positive integers. For $X \geq \sqrt{d}$, let $\mathcal{P}_{A,B,d}(X)$ be the set of positive integer solutions of*

$$Ap^2 - Bq^2 = d, \quad (q, d) = 1, \quad p\sqrt{A} + q\sqrt{B} \leq X.$$

Put

$$R_d(A, B) = \#\{z \pmod{d} : Az^2 \equiv B \pmod{d}\},$$

with $R_1(A, B) = 1$. Then

$$|\mathcal{P}_{A,B,d}(X)| \leq R_d(A, B) \left(1 + \left\lfloor \log_{2+\sqrt{3}} \frac{X}{\sqrt{d}} \right\rfloor\right) \leq d \left(1 + \left\lfloor \log_{2+\sqrt{3}} \frac{X}{\sqrt{d}} \right\rfloor\right). \quad (40)$$

In particular, the orbit count depends on the determinant d , not on the possibly growing product AB .

Proof. Every solution determines

$$z \equiv pq^{-1} \pmod{d}, \quad Az^2 \equiv B \pmod{d}.$$

Take two solutions (p_1, q_1) and (p_2, q_2) with the same z , and put

$$u = \frac{Ap_1p_2 - Bq_1q_2}{d}, \quad v = \frac{p_1q_2 - p_2q_1}{d}.$$

The common-root congruence shows that both numerators are divisible by d . The identity

$$(Ap_1p_2 - Bq_1q_2)^2 - AB(p_1q_2 - p_2q_1)^2 = (Ap_1^2 - Bq_1^2)(Ap_2^2 - Bq_2^2) = d^2$$

therefore gives

$$u^2 - ABv^2 = 1.$$

Writing $\alpha_i = p_i\sqrt{A} + q_i\sqrt{B}$ and $\beta_i = p_i\sqrt{A} - q_i\sqrt{B}$, one also has

$$\frac{\alpha_1}{\alpha_2} = u - v\sqrt{AB}.$$

Here $\alpha_i > 0$, $\beta_i = d/\alpha_i > 0$, and the conjugate of the displayed quotient in $\mathbb{Q}(\sqrt{AB})$ is β_1/β_2 . Hence, after ordering the two solutions by α_i , the larger of a nontrivial quotient and its inverse is a totally positive unit greater than 1. If AB is a square, the displayed unit equation forces $v = 0$ and $u = 1$, so the two solutions coincide. Otherwise every nontrivial integral totally positive unit greater than 1 is at least $2 + \sqrt{3}$: indeed it has the form $x + y\sqrt{AB}$ with $x \geq 2$, $y \geq 1$, and $AB y^2 = x^2 - 1 \geq 3$.

Thus the solutions in one root class form a geometric sequence of ratio at least $2 + \sqrt{3}$. Since α_i has norm d and is larger than its positive conjugate, $\alpha_i \geq \sqrt{d}$. Each root class consequently contains at most the first parenthesis in (40). Summing over the $R_d(A, B) \leq d$ root classes proves the claim. $\square$

Corollary 6.6 (A one-payment two-large-witness tail). *Let $1 \leq b < c \leq N$, put $d = c - b$, and let $Y \geq d$. Define*

$$K = \left\lfloor \frac{N^2 + 1}{Y^2} \right\rfloor, \quad L = 1 + \left\lfloor \log_{2+\sqrt{3}} \left(2\sqrt{\frac{N(N^2 + 1)}{d}} \right) \right\rfloor.$$

The number of $a \leq N$ for which there are primes $p, q > Y$ satisfying

$$p^2 \mid ab + 1, \quad q^2 \mid ac + 1$$

is at most

$$K^2 dL. \tag{41}$$

The same bound holds after interchanging b and c , and after restricting a to any progression.

Proof. For each counted a , choose one pair of witnessing primes and write

$$ab + 1 = p^2 r, \quad ac + 1 = q^2 s.$$

Then $1 \leq r, s \leq K$, while

$$cp^2 r - bq^2 s = c(ab + 1) - b(ac + 1) = d.$$

Since $q > Y \geq d$, one has $(q, d) = 1$. For fixed (r, s) apply Lemma 6.5 with $A = cr$ and $B = bs$. Its height parameter may be taken to be $2\sqrt{N(N^2 + 1)}$, because

$$p\sqrt{cr} = \sqrt{c(ab + 1)} \leq \sqrt{N(N^2 + 1)}$$

and the analogous inequality holds for $q\sqrt{bs}$. Thus each of the K^2 pairs (r, s) contributes at most dL solutions. Forgetting primality and progression conditions only enlarges the count, proving (41). $\square$

The factor d in (41) was paid separately for every quotient pair. For an actual-support argument the quotient sets are sparse, and the root classes should instead be summed before the Pell-orbit payment.

Lemma 6.7 (Determinant square-sieve over quotient supports). *Let $1 \leq b < c \leq N$, $d = c - b$, and $Y \geq d$. Let γ_b, γ_c be positive integers and let*

$$\mathcal{R} \subseteq [1, K_b], \quad \mathcal{S} \subseteq [1, K_c].$$

Consider the integers $a \leq N$ for which there are primes $p, q > Y$ and $r \in \mathcal{R}, s \in \mathcal{S}$ satisfying

$$ab + 1 = \gamma_b p^2 r, \quad ac + 1 = \gamma_c q^2 s.$$

Put

$$A_0 = c\gamma_b, \quad B_0 = b\gamma_c, \quad g_A = (A_0, d), \quad g_B = (B_0, d)$$

and

$$L = 1 + \left\lfloor \log_{2+\sqrt{3}} \left(2\sqrt{\frac{N(N^2 + 1)}{d}} \right) \right\rfloor.$$

Then their number is at most

$$L \sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{S}} R_d^\times(A_0 r, B_0 s) \leq L \min\{d|\mathcal{R}||\mathcal{S}|, |\mathcal{R}|(g_B K_c + d), |\mathcal{S}|(g_A K_b + d)\}. \tag{42}$$

Here $R_d^\times(A, B)$ counts only roots $z \pmod{d}$ with $(z, d) = 1$. This remains valid after restricting a to any progression. In the transformed $q_0 = 25$ progression one may take $\gamma_t \in \{1, 5\}$ and let $\mathcal{R}, \mathcal{S}$ be the exact actual-support/mod-5 quotient masks.

Proof. For a counted a choose one witness pair. Eliminating a gives

$$A_0rp^2 - B_0sq^2 = d.$$

Since $p, q > Y \geq d$, one has $(p, d) = (q, d) = 1$. For fixed (r, s) , Lemma 6.5 therefore gives at most $R_d^\times(A_0r, B_0s)L$ solutions, since the associated root $z = pq^{-1}$ is a unit. Its height bound is unchanged because

$$p\sqrt{A_0r} = \sqrt{c(ab+1)}, \quad q\sqrt{B_0s} = \sqrt{b(ac+1)}.$$

Summing first proves the first inequality in (42).

For the second, interchange the quotient and root sums. There are at most d possible unit roots z , and a root contributes precisely when

$$B_0s \equiv A_0rz^2 \pmod{d}.$$

For fixed (z, r) this linear congruence is either insoluble or is one class modulo d/g_B . Hence among $s \in \mathcal{S} \subseteq [1, K_c]$ it has at most

$$\min \left\{ |\mathcal{S}|, \frac{g_B K_c}{d} + 1 \right\}$$

solutions. Summing over z and r gives both $d|\mathcal{R}||\mathcal{S}|$ and $|\mathcal{R}|(g_B K_c + d)$. Interchanging the two sides gives the third bound because z^2 is a unit, so $(A_0z^2, d) = g_A$. Taking their minimum proves the assertion. Discarding a progression condition only enlarges the count. $\square$

For $Y = \lfloor N/3 \rfloor$, Corollary 6.4 also pays the branch in which exactly one witness exceeds Y . The remaining large-prime obligation is therefore the medium window between the finite sieve cutoff and $N/3$, including its endpoint discrepancy; it is still part of the same distinct-prime debit.

For the pure 18-class parity blocks, the determinant lemma has the following finite very-large specialization.

Corollary 6.8 (Pure parity determinant debit at the half split). *Let $107 \leq N \leq 2 \cdot 10^8$, put $Y = \lfloor N/2 \rfloor$, and let $t_1 < t_2 < t_3$ be members of C_N^{18} of one parity and of span at most 200. The number of members a of the opposite base parity for which at least two of the three forms $at_i + 1$ have prime-square witnesses greater than Y is at most*

$$16 \left(1 + \left\lfloor \log_{2+\sqrt{3}} \left(2 \sqrt{\frac{N(N^2+1)}{50}} \right) \right\rfloor \right). \quad (43)$$

At $N = 2 \cdot 10^6$ this is at most 256; retaining the actual pivot supports lowers the exhaustive maximum to 224.

Proof. For a pure pivot $t \equiv 18 \pmod{25}$ one has

$$at + 1 = p^2r, \quad r \equiv 2 \text{ or } 3 \pmod{5}.$$

The product at is even in the opposite-parity block, so $at + 1$ and hence r are odd. Since $p > Y$ and $N \geq 107$, one has $(tN + 1)/p^2 < 5$. Thus the only possible quotient is $r = 3$; every prime divisor of t may only delete that quotient by its quadratic-residue condition.

Three same-parity members of the 18 class and span at most 200 have one of the six offset patterns in the first column below. For each pair apply Lemma 6.7 with its exact quotient set contained in $\{3\}$. Summing the unit root classes over the three pairs, and then maximizing over the finitely many residues $t_1 \equiv 18 \pmod{25}$ modulo the three gaps, gives

$(t_2 - t_1, t_3 - t_1)$	(50, 100)	(50, 150)	(50, 200)	(100, 150)	(100, 200)	(150, 200)
root-class sum	8	10	14	10	16	14.

Every gap is at least 50, so its orbit factor is at most the parenthesis in (43). A union bound over the three pairs proves the assertion. The residue table and the actual-support scan over all 479960 eligible triples at $N = 2 \cdot 10^6$ are reproduced by `scripts/systematic_pure_determinant_tail.py`. $\square$

There is an even sharper one-form observation at this particular split.

Corollary 6.9 (The pure very-large branch is terminally paid). *Under the hypotheses of Corollary 6.8, the number of opposite-parity base vertices for which at least one of the three forms has a prime-square witness greater than Y is at most 192. Consequently the determinant debit is not the controlling term in the pure half-split continuation.*

Proof. The preceding quotient argument leaves only

$$3p^2 = at + 1.$$

It forces $t > (3(Y + 1)^2 - 1)/N > 3N/4 - 2$. The interval

$$Y < p \leq \sqrt{(tN + 1)/3}$$

has length less than $0.079N < t$. Thus each root class of $3x^2 \equiv 1 \pmod{t}$ contains at most one possible p . If $3 \mid t$ or $4 \mid t$ there are no roots. Otherwise the prime 5 is absent and every odd prime-power factor contributes at most two roots. Since

$$7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 = 215656441 > 2 \cdot 10^8,$$

t has at most six relevant odd prime divisors. Hence there are at most $2^6 = 64$ witnesses for one pivot, and the union over three pivots is at most 192. $\square$

Corollaries 6.8 and 6.9 unconditionally remove the very-large branch from the pure matching obstruction. The remaining medium window $Q < p \leq N/2$ is bounded jointly by separating “at least two small” from “at least two tail” witnesses. Let

$$\mathcal{P}_{47} = \{p : 3 \leq p \leq 47, p \neq 5\}, \quad z_p = p^{-2}.$$

Put

$$\begin{aligned} \lambda_3 &= 1 - e_1(z) + e_2(z) - e_3(z) > 0.847622705884, \\ \alpha_2 &= 1 - 3\{1 - e_1(2z)\} + 2\{1 - e_1(3z) + e_2(3z)\} < 0.109696396217. \end{aligned}$$

The first polynomial is the degree-3 lower Bonferroni bound for one form. For the second, if S_i is the event that form i avoids every finite square class, the indicator that at least two forms have a finite witness is

$$1 - e_2(S_1, S_2, S_3) + 2e_3(S_1, S_2, S_3).$$

The degree-1 lower bound for each two-form survivor and the degree-2 upper bound for the three-form survivor give precisely α_2 . Their respective endpoint budgets are

$$377, \quad 3e_1(21) + 2\{e_1(31) + e_2(31)\} = 1560. \quad (44)$$

Let $Y = \lfloor N/25 \rfloor$ and

$$\Pi(N) = \frac{Y}{\log Y - 1.1} + 1.34, \quad \tau_{47} = 0.003887.$$

For $G > 0$ define

$$T_G(N) = \left(\frac{N}{50} + 1 \right) \tau_{47} + \Pi(N) - 15 + G, \quad (45)$$

$$D_G(N) = \lambda_3 \left(\frac{N}{50} - 1 \right) - 377 - T_G(N), \quad (46)$$

$$U_G(N) = \alpha_2 \left(\frac{N}{50} + 1 \right) + 1560 + \frac{3}{2} T_G(N). \quad (47)$$

The prime bound $\pi(Y) \leq \Pi(N)$ on the range used below follows from (22) for $Y > 60184$; the added 1.34 covers the remaining integer interval $60000 \leq Y \leq 60184$, which is checked directly.

Lemma 6.10 (Joint pure-tail certificate). *For $1.5 \cdot 10^6 \leq N < 2 \cdot 10^8$, every vertex in a pure opposite-parity block has degree at least $D_G(N)$, and every eligible span-200 triple has common non-neighbourhood at most $U_G(N)$, where*

$$G = \begin{cases} 1541.126, & 1.5 \cdot 10^6 \leq N < 2.3 \cdot 10^6, \\ 3605, & 2.3 \cdot 10^6 \leq N < 2 \cdot 10^8. \end{cases}$$

On both intervals

$$D_G(N) > U_G(N), \quad D_G(N) > \frac{N}{125} + 2. \quad (48)$$

Proof. The one-form finite sieve gives the first term of (46). For $47 < p \leq Y$, the square-density is included in τ_{47} and there is one terminal class per prime. For $p > Y$, write $at + 1 = p^2 m$. Since $Y \geq N/26$ and $t \leq N$, one has $m \leq 676$. Opposite parity makes m odd, the mod-5 calculation makes it a nonsquare modulo 5, and every odd prime divisor of t imposes its exact quadratic-residue condition. The transformed root payment is

$$H_t \left(s_t + \frac{2(N/50 + 1)}{Y} \right), \quad \frac{2(N/50 + 1)}{Y} < 1.04004,$$

where s_t is the exact survivor count. Exhausting all 92000 pure pivots $t \leq 2.3 \cdot 10^6$ gives maximum 1541.12512, attained at

$$t = 2038168, \quad \nu_2(t) = 3, \quad H_t = 128, \quad s_t = 11.$$

For the full upper interval, retain the smallest $k - 1$ of the k non-5 odd prime divisors of t . Requiring that one further prime still fit below $2 \cdot 10^8$ gives the following unconditional relaxation:

k	0	1	2	3
s_k	135	135	83	48
H_k	8	16	32	64
$H_k(s_k + 1.04004)$	1088.321	2176.641	2689.282	3138.563

k	4	5	6	7
s_k	26	13	6	1
H_k	128	256	512	1024
$H_k(s_k + 1.04004)$	3461.126	3594.251	3604.501	2089.001

Thus $G = 3605$ is valid. This proves the degree bound.

Now let a be a common non-neighbour of three pivots. If at least two forms have witnesses in $\mathcal{P}_{47}$, the finite polynomial α_2 and the second error in (44) pay it. Otherwise at least two forms have witnesses exceeding 47. If H_i denotes the latter tail event for form i , then

$$\#\{a : \text{at least two } H_i\} \leq \frac{1}{2} \sum_{i=1}^3 |H_i| \leq \frac{3}{2} T_G(N).$$

This proves (47); in particular the medium terminal events are paid jointly rather than independently.

It remains only to check the two scalar margins. Both are increasing on each displayed interval. Between jumps of Y this is immediate; at a jump, differentiating $y/(\log y - 1.1)$ gives an increment less than 0.091, which is dominated by the linear increment in both margins. More explicitly, an increment of 25 in N increases $D_G - U_G$ by more than

$$\frac{\lambda_3 - \alpha_2}{2} - \frac{5}{4} \tau_{47} - 2.5(0.091) > 0.136,$$

and increases $D_G - (N/125 + 2)$ by more than

$$\frac{\lambda_3}{2} - \frac{1}{5} - \frac{1}{2} \tau_{47} - 0.091 > 0.130.$$

At the two left endpoints the rigorous rounded values are

N	G	D_G	U_G	$D_G - U_G$
1500000	1541.126	> 17347.432	< 16406.103	> 941.330
2300000	3605	> 25936.158	< 25621.101	> 315.058

and $D_G - (N/125 + 2)$ is respectively greater than 5345.43 and 7534.15. The support enumeration, finite polynomials, prime counts, and endpoint margins are reproduced by `scripts/systematic_pure_joint_tail_close.py`. $\square$

Theorem 6.11 (Pure matching down to one and a half million). *For every $N \geq 1.5 \cdot 10^6$, the squarefree-edge graph between C_N^{18} and C_N has a matching saturating C_N^{18} . Hence every $B \subseteq C_N^{18}$ satisfies the original Hall inequality.*

Proof. Theorem 5.22 handles $N \geq 2 \cdot 10^8$. In the remaining range work in one parity block and suppose S is a Hall defect, with $m = |S|$. The forward degree bound gives $m > D_G(N) > N/125 + 2$. If no three consecutive members of S had span at most 200, their spans, being multiples of 50, would all be at least 250, whence

$$250(m-2) \leq \sum_{j=1}^{m-2} (x_{j+2} - x_j) \leq 2(N-1),$$

contrary to $m > N/125 + 2$. Choose such a triple.

As in Theorem 5.14, the reverse degree bound and the assumed Hall defect force the unused right-hand set to have more than $D_G(N)$ members. Every one is a common non-neighbour of the chosen triple, contradicting $U_G(N) < D_G(N)$. Hall's theorem supplies the matching in both parity blocks, and restricting it to B proves the last assertion. $\square$

6.1 Joint mixed tail descent to forty million

The same joint split has a stronger mixed form. Averaging the all-three-finite intersection with the two-of-three event preserves the mixed diagonal balance.

Lemma 6.12 (Half finite-intersection, half two-of-three). *For three pairs of events F_i, H_i one has pointwise*

$$\mathbf{1}_{\bigcap_{i=1}^3 (F_i \cup H_i)} \leq \frac{1}{2} \mathbf{1}_{F_1 \cap F_2 \cap F_3} + \frac{1}{2} \mathbf{1}_{\{\#\{i: F_i\} \geq 2\}} + \frac{1}{2} \sum_{i=1}^3 \mathbf{1}_{H_i}. \quad (49)$$

Proof. If exactly k of the F_i occur and all three unions occur, the right side is at least 1 for $k = 3, 2, 1, 0$, respectively: its minimum in the four cases is $1, 1, 1, 3/2$. This also proves the assertion when some F_i and H_i occur simultaneously. $\square$

We apply the lemma with F_i the existence of a prime-square witness through 47 and H_i the existence of a witness above 47. Put

$$\eta = 0.00015, \quad \tau_{47} = 0.003887.$$

Retaining the nonzero square coset modulo 5 in both the finite and actual-support degree arguments gives the following low block.

Proposition 6.13 (Mixed residual below two hundred million). *For $4 \cdot 10^7 \leq N < 2 \cdot 10^8$, every mixed Hall defect satisfies*

$$|A^*| > 0.00030627N > 2\eta N.$$

Proof. First let $4 \cdot 10^7 \leq N < 8 \cdot 10^7$. Use the one-form degree-3 Bonferroni lower bound through 47, split the remaining primes at $Y = \lfloor N/70 \rfloor$, and retain the actual support and worse nonzero mod-5 coset in the root tail. The global support at $8 \cdot 10^7$ fits in $M = 4902$. Evaluated at the lower endpoint, the least row is

$$\text{odd}, \quad k = 4, \quad H = 32, \quad s_k = 286, \quad (p_1, p_2, p_3) = (61, 73, 79).$$

After the reciprocal-survivor correction the normalized mixed residual is greater than

$$0.0077080.$$

For $8 \cdot 10^7 \leq N < 2 \cdot 10^8$, use $D = \lfloor N/300 \rfloor$ and the common box $M = 90001$ exactly as in Proposition 5.20, but intersect every complementary mask with the worse of the two nonzero square cosets modulo 5. To make the calculation uniform, take the prime-support superset at $2 \cdot 10^8$ and evaluate every fixed row at $N = 8 \cdot 10^7$. The least row is

$$\text{odd}, \quad k = 3, \quad H = 16, \quad s_k = 9176, \quad (p_1, p_2) = (449, 491).$$

Its excess over $N/50$ is greater than 12542.5712308. The same reverse degree subtraction, after including the reciprocal-survivor correction < 291.111 , therefore gives

$$\frac{2(12542.5712308 - 291.111) - 1}{8 \cdot 10^7} > 0.00030627.$$

In both blocks the global support is fixed, the normalized affine main term increases, and all normalized positive payments decrease. Thus the lower endpoints control their intervals. The support enumeration is reproduced by `scripts/systematic_mixed_interval_support.py`; the endpoint subtraction is checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

We next record the finite payment. On an odd-prime progression let S_i be the event that the i th form survives every prime through 47. The indicator of at least two finite-bad forms is

$$1 - \sum_{i < j} \mathbf{1}_{S_i S_j} + 2\mathbf{1}_{S_1 S_2 S_3}.$$

Use degree-1 lower bounds for the three pair survivors and a degree-2 upper bound for the triple survivor. At a common prime the pair multiplicities sum to 3 and the triple multiplicity is 1. At a pair-coincident non-common prime they are 5 and 2; without a coincidence they are 6 and 3. If $z_p = p^{-2}$, declaring a non-common prime pair-coincident changes the resulting upper polynomial by

$$z_p(1 - 2S_p) > 0, \quad S_p < 3 \sum_{\substack{3 \leq q \leq 47 \\ q \neq 5}} q^{-2} < \frac{1}{2}.$$

Hence all non-common odd primes may be declared pair-coincident for this upper bound. A common prime only decreases the two-of-three polynomial, so the two finite polynomials are evaluated jointly.

Prime 2 is exact. For a triple meeting both odd classes modulo 4, one mod-4 class makes the close pair automatically bad, the opposite class makes the third form automatically bad, and the other two classes use the odd-prime two-of-three polynomial. For three pivots in one odd mod-4 class, one class makes all three forms automatically bad and the other three use the odd-prime polynomial.

If the even mass exceeds ηN , a close pair in one even mod-4 class has gap less than 13334. After charging the even mass in the other case, the odd mass is greater than $0.0001635517807N$. A close pair in the denser of two odd classes has gap less than 12229; if only one odd class occurs, its close pair has gap less than 6115. For any common odd-prime product Q , the close pair gives $4Q^2 < d$. Exhausting the resulting cutoff-prime supports gives

branch	maximizing common support	$I_3 + I_2$	$\epsilon_3 + \epsilon_2$
even generic	7	< 0.223675978	4874
even concentrated	3, 7	< 0.309830838	4510
two odd classes	$\emptyset$	< 0.541995982	7113
two odd, common 3	3	< 0.543144010	6615
one odd class	2	< 0.679312680	7435
one odd, common 2, 3	2, 3, 7	< 0.739301322	6423.

Here I_3 is the degree-3/2/3 all-three-finite bound used above, and I_2 is the new two-of-three bound. By Lemma 6.12, the normalized finite payment across the two base progressions is therefore

$$\frac{I_3 + I_2}{25} + \frac{\epsilon_3 + \epsilon_2}{N}. \quad (50)$$

The finite polynomial (50) is combined with the ten-branch envelopes below, each of which includes the reciprocal mass beyond 3025. This keeps each root bound paired with the finite row at the same cutoff.

Finally, the eventwise Möbius–CRT diagonal scan with $R = 2 \cdot 10^7$ uses the following entirely rational tail estimate. If P is the exact truncated signed count at the right endpoint U , and θ is the density of the permitted residue set, then throughout $[L, U]$

$$\frac{\Delta(N)}{N} < \frac{1}{L} \left\{ P + \frac{38\theta U}{R} + \frac{13(U^2 + 1)}{R^2} \right\}. \quad (51)$$

Indeed $\log(2 \cdot 10^7) + 2 < 19$: the positive Taylor series gives $e > 2.7$, while $2.7^{17} > 2 \cdot 10^7$. Every comparison in (51) is made by unsigned-integer cross-multiplication. On all nine blocks from $5 \cdot 10^6$ to $2 \cdot 10^8$ this gives the uniform strict ceilings

support	all	even cell	odd or 8 a	E_2 cell
$10^6 \Delta / N <$	25294	19681	15812	16164
support	odd union	odd+cell	one odd	one odd cell
$10^6 \Delta / N <$	12652	7036	6330	711.

The parameterized scan is reproduced by

`scripts/systematic_concentrated_diagonal_blocks.cpp.`

Theorem 6.14 (Joint mixed close to forty million). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every*

$$N \geq 4 \cdot 10^7.$$

Proof. Theorem 5.23 handles $N \geq 2 \cdot 10^8$. In the remaining interval take a mixed defect and put $A = B \cup T_N(B)$. Proposition 6.13 supplies more than three sparse-class charges. Apply the ten exhaustive valuation/parity alternatives used in Proposition 6.18; their selection proof uses only the residual inequality and the gap lemma, and is therefore valid unchanged on this larger interval.

For the six blocks

$$[40, 50), [50, 70), [70, 80), [80, 100), [100, 150), [150, 200) \quad \text{million,}$$

the transformed-root ceilings for E_1, E_2, E_3 , and the odd classes are

$[L, U)$	E_1	E_2	E_3	O
[40, 50)M	.009677836	.009973185	.010991872	.010396089
[50, 70)M	.009350108	.009648325	.010405463	.009926965
[70, 80)M	.008921868	.009135839	.009679089	.009335765
[80, 100)M	.008771555	.008959044	.009511773	.009134226
[100, 150)M	.008534837	.008684996	.009127671	.008825297
[150, 200)M	.008175215	.008275070	.008569448	.008368371.

These ceilings include the reciprocal-survivor correction. Combine them with the exact diagonal grid, the ten finite payments

$$.008685, .012616, .019420, .020878, .026643, .029459$$

in their respective generic/common branches, the sparse-class charge $1/20001 + 1/L$, and the cutoff-23 or cutoff-19 square tail. Exact rational comparison shows that E_2 -generic is the largest row in every block:

$[L, U)$	$\max A /N$	$1/25 - 7/(25L) - \max A /N$
[40, 50)M	.039972160176	$> .0000278328$
[50, 70)M	.039810601176	$> .0001893932$
[70, 80)M	.039554376176	$> .0004456198$
[80, 100)M	.039466546676	$> .0005334498$
[100, 150)M	.039330914676	$> .0006690825$
[150, 200)M	.039127498676	$> .0008724994.$

For example, with $\widehat{\tau}_{23}$ as in the ten-million proof, the controlling lower-endpoint total is

$$C_{40} = \frac{25288306}{10^9} + \frac{8685}{10^6} + \frac{3}{25} \widehat{\tau}_{23} + \frac{1}{2} \frac{9973185}{10^9} < \frac{1}{25} - \frac{7}{25(4 \cdot 10^7)} - \frac{27}{10^6}.$$

Thus every branch contradicts the Hall defect. The rows and exact rational inequalities are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

6.2 Valuation, cell, and fibre descent to five million

The remaining factor-of-eight descent refines the same mixed defect by the 2-adic valuation of its even pivots and by the actual residue cells occupied by the odd pivots. We first isolate the two finite combinatorial facts used below.

Lemma 6.15 (Valuation support and the four-pivot inequality). *Let a be even, p odd, and $p^2 m = at + 1$. If $\nu_2(a) = 1, 2$, or at least 3, then respectively*

$$m \equiv 1 \pmod{2}, \quad m \equiv 1 \pmod{4}, \quad m \equiv 1 \pmod{8}.$$

Moreover, for four pairs of events F_i, H_i ,

$$\mathbf{1}_{\cap_{i=1}^4 (F_i \cup H_i)} \leq \frac{1}{2} \sum_{|S|=3} \mathbf{1}_{\cap_{i \in S} F_i} + \frac{1}{2} \sum_{i=1}^4 \mathbf{1}_{H_i}. \quad (52)$$

More generally, for n pairs and $1 \leq k \leq n$,

$$\mathbf{1}_{\cap_{i=1}^n (F_i \cup H_i)} \leq \mathbf{1}_{\#\{i: F_i\} \geq k} + \frac{1}{n - k + 1} \sum_{i=1}^n \mathbf{1}_{H_i}. \quad (53)$$

Proof. Every odd square is 1 modulo 8. Since $2^{\nu_2(a)} \mid at$ and $p^2 \equiv 1 \pmod{8}$, division by p^2 gives the three displayed quotient restrictions. For (52), if exactly j of the F_i occur, the right side is at least $2, 3/2, 1, 1, 2$ for $j = 4, 3, 2, 1, 0$, respectively. For (53), either at least k finite events occur, or at least $n - k + 1$ of the unions must be supplied by their H_i terms. $\square$

Lemma 6.16 (Two-coordinate fibres and capacitated cells). *Let X be a finite set and let $r_q : X \rightarrow \mathbb{Z}/q^2\mathbb{Z}$.*

1. *If every pair of elements of X agrees in at least one of the two coordinates r_7, r_{11} , then X is contained in one mod-49 fibre or in one mod-121 fibre.*
2. *Let S be a set of s occupied mod-9 cells, joined to the mod-49 residues which they contain, and give every right vertex capacity two. If there is no matching of size eight, then some $Y \subseteq S$ satisfies*

$$|Y| - 2|N(Y)| \geq s - 7. \quad (54)$$

Proof. For the first assertion, choose two points with different first coordinates. They have the same second coordinate. Any third point which had a different second coordinate would have to share both distinct first coordinates, which is impossible. The second assertion is the max-flow/min-cut formula for a bipartite matching with right capacities two:

$$\nu = \min_{Y \subseteq S} (s - |Y| + 2|N(Y)|).$$

If $\nu \leq 7$, rearranging gives (54). $\square$

Lemma 6.17 (Gap selection and cell charge). *Let $X \subseteq [1, N]$ and let $G \geq 1$. If no two members of X have difference at most G , then*

$$|X| \leq \frac{N-1}{G+1} + 1 < N \left(\frac{1}{G+1} + \frac{1}{N} \right).$$

Consequently, if X is split into r residue cells, each nondense cell of cardinality at most $(N-1)/(G+1) + 1$ may be discarded at total normalized cost at most

$$r \left(\frac{1}{G+1} + \frac{1}{N} \right).$$

Proof. If $x_1 < \dots < x_m$ are the members of X , then $x_{i+1} - x_i \geq G+1$, so $(m-1)(G+1) \leq N-1$. Summing this estimate over the cells proves the second assertion. $\square$

We now close the two upper intervals. In both proofs split the even pivots into E_1, E_2, E_3 according as $\nu_2 = 1, 2$, or at least 3. The quotient restrictions in Lemma 6.15 reduce their transformed root supports.

Proposition 6.18 (Valuation close on the twenty-million block). *The Hall inequality (2) holds for*

$$2 \cdot 10^7 \leq N < 4 \cdot 10^7.$$

Proof. The cutoff-19 degree-3 one-form certificate gives

$$|A^*| > 0.006910733259914470N.$$

Put

$$\delta_{20} = \frac{1}{20001} + \frac{1}{2 \cdot 10^7}.$$

The four six-form root ceilings for E_1, E_2, E_3 , and an odd pivot are

$$.008622089, \quad .010006474, \quad .012915119, \quad .010177869.$$

If $|E_1| > \delta_{20}N$, Lemma 6.17 supplies a pair of gap at most 20000. Add a third E_1 pivot. If the class is nonconstant modulo 9, choose the third pivot outside the pair's cell and use the generic row; otherwise use the common-3 row. This is the three-pivot inequality (49); no four-pivot claim is used in this interval. If $|E_1| \leq \delta_{20}N$, charge it and repeat with E_2 . If both have been charged, repeat with E_3 ; the off-base diagonal is then supported on the odd-or-divisible-by-8 set. If all three even classes have been charged, their total cost is at most $3\delta_{20}N$.

The remaining odd mass is greater than

$$(0.006910733259914470 - 3\delta_{20})N.$$

If both odd mod-4 classes occur, its denser class contains a pair of gap less than twice the reciprocal of this density; add a pivot from the other class. If only one odd class occurs, its close-pair bound is the reciprocal of the same density and the third pivot is taken in that class. Splitting each alternative according as the three pivots are constant modulo 9 gives the final four rows. Thus the following ten rows are exhaustive:

row	E_1G	E_{13}	E_2G	E_{23}	E_3G
total <	.039257774	.037068015	.039949967	.034297022	.032021074

row	E_{33}	O_2G	O_{23}	O_1G	O_{13}
total <	.035449069	.039924070	.034577881	.039438565	.036137568.

The controlling row is now the generic E_2 row. With the exact finite and square-tail payments, its total is

$$C_{20} = \frac{25289550}{10^9} + \frac{1572362642548502371982744227834515457}{171081196711186531200844968164610000000} \quad (55)$$

$$+ \frac{11661}{25000000} + \frac{1}{2} \frac{10006474}{10^9}. \quad (56)$$

Exact common-denominator arithmetic gives

$$C_{20} < \frac{1}{25} - \frac{7}{25(2 \cdot 10^7)} - \frac{50}{10^6}.$$

All other rows have larger slack, and every normalized endpoint payment decreases while the Hall target increases. This proves the full interval. The support exhaustion is reproduced by the accompanying finite enumerations, and the ten rational totals are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

Proposition 6.19 (Cell close on the ten-million block). *The Hall inequality (2) holds for*

$$10^7 \leq N < 2 \cdot 10^7.$$

Proof. Call a mod-9 cell dense if its cardinality is more than $(N - 1)/1000001 + 1$, and put

$$\delta_{10} = \frac{1}{1000001} + \frac{1}{10^7}.$$

The cutoff-17 degree certificate gives $|A^*| > .005826196348744825N$. The relevant root ceilings are

$$.010545872, \quad .011845571, \quad .014894092, \quad .013231580.$$

In E_1 , two dense cells supply two close pairs and hence the four-pivot row (52); one dense cell supplies three pivots and the other eight cells cost at most $8\delta_{10}N$. If there is no dense cell, retain the whole class in the unrestricted diagonal and continue to E_2 . The same alternatives for E_2 give the unrestricted two-cell row or, after charging the nine E_1 cells and eight unused E_2 cells, the E_2 -cell row with charge $17\delta_{10}N$. If neither class supplies a row, charge both and use a dense E_3 cell; this costs $18\delta_{10}N$ and leaves the odd-or-divisible-by-8 diagonal. If no even class supplies a row, all 27 even cells cost at most $27\delta_{10}N$.

For the odd remainder use the cutoff-7 threshold allocation described below for the five-million block. Discarding odd cells of size at most five costs at most 90 pivots. The alternatives are: disjoint triples of active cells in the two odd mod-4 classes; at most eleven active cells; one concentrated residue; or one odd mod-4 class with $s = 1, \dots, 9$ active cells. Lemma 6.16 and the exact recurrence (61) cover the $s = 8, 9$ matching failures. This avoids the invalid inference that crossed common supports at 7 and 11 must put all pivots into one fixed prime-square fibre.

Using the five-million diagonal ceilings, which also dominate the direct scan on this interval, every odd row is at most 0.039247000. The five even rows are, respectively,

row	E_1 two	E_1 one	E_2 two	E_2 one	E_3
total <	.039077928	.039259999	.039944394	.036406507	.037579342.

Thus the controlling row is E_2 with two dense cells. Its four three-form finite intersections admit the conservative cutoff-23 density

$$I_{10} = \frac{5355507302216}{165901428816123}, \quad \epsilon_{10} = 607.$$

Put also

$$\widehat{\tau}_{23} = \frac{64081802747648035629863}{7596668444022826249000000}.$$

This upper bound uses cutoff 23. The decimal-free controlling total is

$$C_{10} = \frac{25289862}{10^9} + \frac{4I_{10}}{25} + \frac{4\epsilon_{10}}{10^7} + \frac{4}{25}\widehat{\tau}_{23} + \frac{2}{3}\frac{11845571}{10^9}. \quad (57)$$

Direct common-denominator arithmetic yields

$$C_{10} < \frac{1}{25} - \frac{7}{25 \cdot 10^7} - \frac{55}{10^6}. \quad (58)$$

The exact rational calculation above is reproduced by

`scripts/verify_four_range_paper_arithmetic.py.`

The diagonal ceilings come from

`scripts/systematic_concentrated_diagonal_blocks.cpp.`

□

For the lower block the support constraints are retained jointly. On $5 \cdot 10^6 \leq N < 10^7$, the cutoff-17 degree calculation gives

$$|A^*| > 0.003997632216030162N. \quad (59)$$

The relevant cutoff root ceilings are

$$T_{E_1,23} < .013878159, \quad T_{E_2,7} < .015472245, \quad T_{E_3,23} < .020270750, \quad T_{O,7} < .017398287. \quad (60)$$

For E_1 , call a paired mod-9 cell good if it is contained in neither one mod-49 fibre nor one mod-121 fibre. Lemma 6.16(1) supplies a pair which breaks both primes. Otherwise label the cell 7-only, 11-only, or both. Every cutoff-23 common support compatible with the label is enumerated; in particular, the primes 13, 17, 19, 23 are not silently declared non-common. If there are two good cells, use four pivots. If there is one, pair it with the best bad cell. If there is none, every 7-only or both cell is inserted into the eventwise diagonal through its actual mod-49 fibre. The maxima for $E_{1,0}$ plus $k = 0, \dots, 9$ such fibres have the uniform strict rational ceiling

$$.0191113,$$

instead of the incompatible raw-fibre/unrestricted hybrid. The one-paired cell case uses three pivots. The resulting four E_1 envelopes are

$$.039267556, \quad .039360535, \quad .035990495, \quad .039569683.$$

For E_2 , the two-cell and one-cell rows are below .037494782 and .038211102. If E_1, E_2 are both charged, nineteen E_3 pivots force three in one mod-9 cell; its envelope is below .039335952.

For the odd part, discard every mod-9 cell which contains at most five pivots and call the other cells active. Together with the preceding even charges this costs at most 128 pivots. If both odd mod-4 classes admit disjoint triples of active mod-9 cells, choose three pivots from each. Otherwise an elementary three-set check gives at most eleven active odd cells: if one class has at most two cells the assertion is immediate, while if both have at least three, the failure of disjoint triples forces both cell sets to have size at most five. A pair from each class in different mod-9 cells then gives a four-pivot row below .037135620; if all active cells have the same mod-9 residue, the concentrated three-pivot row is below .033200482.

For one odd mod-4 class, let s be its number of active mod-9 cells. When $s = 8$ or 9 , give the mod-49 residues capacity two. A matching of size eight selects eight pivots with no mod-49 collision block larger than two. If it fails, (54) restricts the diagonal. The eventwise maxima in the two failure cases are

$$.003572, \quad .003587.$$

When $s = 1$, either four mod-49 fibres each supply a pair, or all but at most 49 pivots lie in three such fibres; the latter diagonal is below .000050. The cases $2 \leq s \leq 7$ use six pivots distributed as evenly as possible among the active cells.

Here the finite calculation is genuinely finite, with no hidden limiting step. Put $M = 4 \cdot 9 \cdot 49 = 1764$ and let $V_p(S)$ be the worst conditional probability, after the primes above p have been exposed, that the terminal set of finite-bad forms has the required cardinality. If the collision partition at p is $\mathcal{Q}$, then backwards induction uses

$$V_{p^-}(S) = \max_{\mathcal{Q}} \left(\frac{p^2 - |\mathcal{Q}|}{p^2} V_p(S) + \frac{1}{p^2} \sum_{Q \in \mathcal{Q}} V_p(S \cup Q) \right). \quad (61)$$

Allowing the maximizing partition to depend on S only enlarges the true density. If the resulting density is R/M , then in either base progression every prefix has excess at most

$$R \left(1 - \frac{R}{M} \right), \quad (62)$$

because a prefix of length u contains at most $\min(u, R)$ favourable residues. Thus both the main density and the integer endpoint are certified.

The controlling periodic rows are

branch	R/M	tail factor	diagonal	total
$O_1, s = 2$	40/147	2/3	.001415	.039866076
$O_1, s = 7$	55/196	1/2	.004928	.039932809
$O_1, s = 9, \text{ matching}$	1/4	8/15	.006330	.039701523
$O_1, s = 9, \text{ failure}$	55/196	1/2	.003587	.038591809
$O_2, \text{ balanced}$	1/42	1	.012652	.039374127.

The controlling row admits a decimal-free exact comparison. Define

$$\hat{\tau}_7 = \frac{3887}{10^6} + \sum_{p \in \{11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}} \frac{1}{p^2}.$$

The cutoff-7 dynamic programme gives exactly $R/M = 55/196$ and prefix excess 69795/196. With the upward rational ceilings

$$D_{5,7} < \frac{4928}{10^6}, \quad T_{O,7} \leq \frac{17398287}{10^9},$$

the complete $O_1, s = 7$ row is at most

$$C_5 = \frac{4928}{10^6} + \frac{2}{25} \frac{55}{196} + \frac{2}{5 \cdot 10^6} \left(\frac{55}{196} + \frac{69795}{196} \right) + \frac{1}{2} \frac{117398287}{10^9} + \frac{3}{25} \hat{\tau}_7 + \frac{128}{5 \cdot 10^6}. \quad (63)$$

Exact common-denominator arithmetic yields

$$C_5 < \frac{1}{25} - \frac{7}{25(5 \cdot 10^6)} - \frac{67}{10^6}.$$

Consequently the controlling five-million row retains rational slack greater than $6.7 \cdot 10^{-5}$ after outward rounding.

Every omitted subrow is smaller. The exact rational dynamic programme, prefix payments, root jumps, fibre classifications, and all diagonal maxima are reproduced by

`scripts/systematic_mixed_cell_5m_close.py,`
`scripts/systematic_concentrated_diagonal_blocks.cpp.`

Proposition 6.20 (Cell–fibre close on the five-million block). *The Hall inequality (2) holds for*

$$5 \cdot 10^6 \leq N < 10^7.$$

Proof. Suppose that a strict Hall defect exists. The residual estimate (59) supplies more than $0.003997632216030162N$ pivots. Call an E_1 mod-9 cell paired when it contains at least two pivots. If at least two cells are paired, classify them as good, 7-only, 11-only, or both. Two good cells, one good cell, and no good cell give respectively the first three E_1 envelopes displayed above; Lemma 6.16(1) proves that these cases exhaust the possible 7–11 collision patterns. If exactly one cell is paired and contains at least three pivots, the four one-cell rows apply. Otherwise $|E_1| \leq 10$, which is charged.

After that charge, the identical paired-cell split in E_2 gives its two-cell or one-cell row. If neither applies, $|E_2| \leq 10$ and it too is charged. Nineteen remaining E_3 pivots force three into one of the nine mod-9 cells; if there are fewer, $|E_3| \leq 18$. Thus failure to select an even row costs at most $10 + 10 + 18 = 38$ pivots.

In the odd part, cells containing at most five pivots cost at most $2 \cdot 9 \cdot 5 = 90$ more. Hence the total raw charge is at most 128. If both odd mod-4 classes have active cells and admit disjoint triples of cell residues, use the balanced six-pivot row. If not, the argument above gives at most eleven active cells. Two different residues give the four-pivot row, while one residue gives the concentrated three-pivot row. These are all possibilities with both odd classes active.

It remains that only one odd mod-4 class is active. Let s be its number of active cells. For $2 \leq s \leq 7$, six pivots distributed as evenly as possible give the listed periodic rows. For $s = 8, 9$, a capacity-two matching either selects eight pivots with collision blocks of size at most two or Lemma 6.16(2) gives the displayed restricted diagonal. For $s = 1$, either four mod-49 fibres supply pairs, or at most three fibres are nonsingletons and the remaining at most 49 pivots are charged. Therefore the branch list used in `scripts/systematic_mixed_cell_5m_close.py` is exhaustive.

For every branch, (53), the exact periodic recurrence (61), the prefix bound (62), and the appropriate diagonal and root ceilings bound the Hall completion by its listed total. The largest is the $O_1, s = 7$ row, and (63) puts it strictly below $1/25 - 7/(25N)$. Hence $|B| + |T_N(B)| \leq |C_N|$, contradicting the strict Hall defect. $\square$

Theorem 6.21 (Direct interval splice at five million). *The Hall inequality (2), and hence the exact upper bound in Erdős Problem 848, holds for every*

$$N \geq 5 \cdot 10^6.$$

Proof. Proposition 6.20 handles $5 \cdot 10^6 \leq N < 10^7$, Proposition 6.19 handles $10^7 \leq N < 2 \cdot 10^7$, Proposition 6.18 handles $2 \cdot 10^7 \leq N < 4 \cdot 10^7$, and Theorem 6.14 handles $N \geq 4 \cdot 10^7$. The four intervals are disjoint and exhaustive. $\square$

6.3 Full-support continuation below five million

The low degree calculation can be strengthened because every prime divisor of a pivot, rather than only its smallest prefix, may be retained in the quadratic-residue mask. For $M = 4902$ the exact survivor maxima at product bound $3 \cdot 10^6$ are

	$k = 0$	1	2	3	4	5	6
odd	1961	1194	612	312	153	73	28
$\nu_2 = 1$	981	594	312	156	80	38	15
$\nu_2 = 2$	491	312	171	92	54	28	13
$\nu_2 \geq 3$	246	169	102	58	37	22	12.

These maxima include the worse nonzero square coset modulo 5 and the quotient congruences modulo 2, 4, 8. For example, the controlling support is (73, 149, 211) in the odd $k = 3$ row.

Proposition 6.22 (Full-support low residual and bounded clusters). *On $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, every mixed Hall defect has more than*

$$\alpha N, \quad \alpha = 0.000249932448083113 \dots \quad (64)$$

non-base residual pivots. Among them there are three of span at most 8045 and eight of span at most 28540.

Proof. Put $Y = \lfloor N/70 \rfloor$, $X = N/25 + 1$, and sieve through 17. For each displayed support row retain the active degree-3 Bonferroni lower bound, subtract

$$X\tau_{17} + \pi(Y) - \pi(17) + H \left(s + \frac{2X}{Y} \right) + \frac{N}{50},$$

including its exact finite discrepancy. The normalized minimum is the odd $k = 3$, $H = 16$, $s = 312$ row at $N = 1.5 \cdot 10^6$; before division by N its reverse-degree residual is

$$374.898672124669531 \dots$$

For fixed Y every row is affine in $1/N$ and increases. Between consecutive prime jumps the normalized positive payments decrease; exact rational checks at every prime jump show that the same lower endpoint is the global minimum.

If $x_1 < \dots < x_m$ are the residual pivots, then for $r = 3$ or 8,

$$\sum_{i=1}^{m-r+1} (x_{i+r-1} - x_i) \leq (r-1)(N-1).$$

Using $m > \alpha N$ and exhausting the fewer than 376 possible integer values of $\lfloor \alpha N \rfloor$ gives the two stated spans. The support search, rational degree rows, prime jumps, and integer span exhaustion are reproduced by `scripts/systematic_full_support_scan.cpp`, `scripts/systematic_one_prime_support_scan.cpp`, and `scripts/systematic_full_support_degree_low.py`. $\square$

We use the following exact one-form tail.

Lemma 6.23 (Exact complete one-form tail). *For $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, $1 \leq t \leq N$, and*

$$\mathcal{M}_t(N) = \left\{ a \leq N : \begin{array}{l} a \equiv 7 \text{ or } 18 \pmod{25}, \text{ and some prime } p \text{ satisfies} \\ D < p \leq N, \quad p^2 \mid at + 1 \end{array} \right\},$$

one has

$$|\mathcal{M}_t(N)| \leq 40. \tag{65}$$

Proof. Every event on the stated interval embeds in the upper box $N_* = 3 \cdot 10^6$. Writing $at + 1 = p^2 r$, every event occurs in the finite list

$$D < p \leq N_*, \quad 1 \leq r \leq \left\lfloor \frac{N_*^2 + 1}{p^2} \right\rfloor, \quad p^2 r - 1 = at.$$

There are 213713 primes and exactly 27898239 pairs (p, r) in this list. For each pair, factor $p^2 r - 1$ and retain every divisor $a \leq N_*$ with $a \equiv 7, 18 \pmod{25}$ whose complementary divisor t is also at most N_* . This is an equivalence, not a relaxation. Moreover $D^4 > N_*^2 + 1$, so two different retained primes cannot represent the same pair (a, t) . Exact counters over $1 \leq t \leq N_*$ have maximum

$$40 \quad \text{at } t = 10920.$$

The prime sieve, complete quotient range, all 3063864 retained divisor events, deterministic 64-bit factorization with factor-product verification, and divisor enumeration are implemented by

`scripts/systematic_one_form_medium_exhaustion.cpp`.

Since every smaller- N event is in the enumerated upper box, (65) follows. $\square$

The resulting uniform one-form bounds are as follows.

Lemma 6.24 (Low one-form incidence envelopes). *For $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, put*

$$E(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lfloor \frac{N}{25p^2} \right\rfloor + 182 + 40,$$

$$O(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lfloor \frac{N}{100p^2} \right\rfloor + 268 + 40,$$

where the sums are over primes. If t is even and non-base, then

$$\#\{a \leq N : a \equiv 7, 18 \pmod{25}, \text{ at } + 1 \text{ is nonsquarefree}\} \leq E(N). \tag{66}$$

If t is odd and non-base and $r \pmod{4}$ is fixed, then the number of such $a \equiv r \pmod{4}$ for which $at + 1$ has an odd-prime-square divisor is at most $O(N)$. At the lower endpoint,

$$E(1.5 \cdot 10^6) = 19694, \quad O(1.5 \cdot 10^6) = 5224.$$

Proof. For a fixed odd prime $p \neq 5$, the congruence $p^2 \mid at + 1$ has at most one a -class modulo p^2 . Intersecting it with the two base classes gives two progressions modulo $25p^2$, and imposing one class modulo 4 gives two progressions modulo $100p^2$. These are the two ceiling sums.

On the complete upper box $a, t \leq 3 \cdot 10^6$, exhaustive modular inversion over $300 < p \leq 28540$ gives total prime-incidence maximum 268, and maximum 182 when t is even and non-base. The latter is attained at $t = 120$. Lemma 6.23 supplies the final 40 for $p > 28540$. If t is even, prime 2 is inactive. Prime 5 is always inactive for a non-base pivot: from $25 \mid at + 1$ and $a \equiv 7$ or $18 \pmod{25}$ one obtains respectively $t \equiv 7$ or $18 \pmod{25}$. This proves both bounds. The upper-box enumeration and its exhaustive check are reproduced by `scripts/systematic_low_high_prime_degree.cpp`. $\square$

Lemma 6.25 (First-low parity diagonal ceilings). *On $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, the nonsquarefree diagonal among the non-base classes has the following uniform upper bounds:*

<i>allowed support</i>	<i>count</i>
<i>unrestricted</i>	$0.025297N$
<i>the two odd classes modulo 4</i>	$0.012657N$
<i>one odd class modulo 4</i>	$0.006337N$.

Proof. The eventwise Möbius–CRT diagonal enumeration on 139 blocks applies (51) by exact integer cross-multiplication. Decimal renderings of the largest rational envelopes are

$$0.0252968815411\dots, \quad 0.0126562333156\dots, \quad 0.0063363175457\dots$$

They occur respectively on the blocks beginning at 1665621, 1812996, and 1724796. The square-free semigroup enumeration has 1237599 nodes, 4134027 root terms, and 77041 retained events; the harmonic and negative-Pell tails are then added monotonically at the upper block end-point. The complete producer and the three asserted upward decimal ceilings are in `scripts/systematic_concentrated_diagonal_blocks.cpp`. $\square$

Theorem 6.26 (Direct close of the first low block). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every*

$$1.5 \cdot 10^6 \leq N < 3 \cdot 10^6.$$

Proof. Theorem 6.11 closes the pure opposite-base case. Suppose therefore that a mixed Hall defect exists, put $A = B \cup T_N(B)$, and let A^* be its non-base residual part. By Proposition 6.22, the residual pivot set contains more than

$$0.000249932448083113N > 374$$

members throughout the interval. Every base member of A is a common non-neighbour of every selected residual pivot, while every member of A^* belongs to the nonsquarefree diagonal. We use these two facts in three exhaustive parity branches.

If there are at least eight even residual pivots, select eight of them. For each common base non-neighbour, all eight associated forms are nonsquarefree. Averaging their eight indicators and then taking the union bound over prime-square witnesses gives the common-base count at most $E(N)$. Lemma 6.25 therefore gives

$$|A| \leq 0.025297N + E(N).$$

Otherwise charge the at most seven even residual pivots. Suppose first that both odd classes modulo 4 contain at least eight residual pivots, and select four from each class. Split the base variable a into its four classes modulo 4. When a is even, prime 2 witnesses none of the eight forms. When a is odd, it witnesses exactly one of the two four-pivot classes, so the other four forms still require odd-prime-square witnesses. Averaging the required four or eight indicators on each a -class and applying Lemma 6.24 gives common-base count at most $4O(N)$. Hence

$$|A| \leq 0.012657N + 7 + 4O(N).$$

Finally, one odd pivot class has at most seven members. Charge it together with the at most seven even pivots and select eight pivots from the remaining odd class. On exactly one odd a -class modulo 4, prime 2 witnesses all eight forms; that class contains at most $2\lceil N/100 \rceil$ base values. On the other three a -classes prime 2 is inactive for all selected pivots, so the averaged odd-prime count is at most $3O(N)$. Thus

$$|A| \leq 0.006337N + 14 + 2 \left\lceil \frac{N}{100} \right\rceil + 3O(N).$$

Each right side is strictly smaller than $N/25 - 7/25$ on the full interval. Indeed it is affine and increasing between ceiling jumps, so it suffices to check the lower endpoint and the points immediately following a multiple of $25p^2$, $100p^2$, or 100. The exact scan of all 21999 such points gives

branch	worst N	count margin
eight even	1500751	2359.261953
two odd classes	1500401	20106.184543
one odd class	1500401	4799.718863.

This finite arithmetic is reproduced by `scripts/systematic_low_parity_close.py`. Every branch contradicts the assumed Hall defect, proving the theorem. $\square$

Lemma 6.27 (Second-low support and one-form envelopes). *For $3 \cdot 10^6 \leq N < 5 \cdot 10^6$, every mixed Hall defect has more than*

$$0.00157645N \tag{67}$$

non-base residual pivots. Put

$$E_5(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lceil \frac{N}{25p^2} \right\rceil + 278 + 49,$$

$$O_5(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lceil \frac{N}{100p^2} \right\rceil + 383 + 49,$$

where the sums are over primes. If t is even and non-base, its nonsquarefree degree into the two base classes is at most $E_5(N)$. If t is odd and non-base and one class of a modulo 4 is fixed, its odd-prime-square degree into that class and the two base classes is at most $O_5(N)$. At the lower endpoint,

$$E_5(3 \cdot 10^6) = 39211, \quad O_5(3 \cdot 10^6) = 10192.$$

Proof. For the residual assertion, repeat the reverse-degree calculation of Proposition 6.22 with upper support $5 \cdot 10^6$, cutoff 17, split 70, and support modulus 4902. For each possible number $0 \leq k \leq 6$ of retained prime divisors and each 2-adic type, take independently the smallest degree-3 Bonferroni density, the largest endpoint error, and the largest full-support quadratic-residue survivor count. This enlarges every actual debit. On a fixed cell $Y = \lfloor N/70 \rfloor$, the resulting normalized residual has the form $A + B/N$, so its minimum is at a cell endpoint. Exact rational exhaustion of all cells and all 28 relaxed rows has minimum

$$0.001576458682760064\dots > 0.00157645$$

at $N = 3000130$. The controlling relaxed row is odd, has $k = 3$, root factor 16, and survivor maximum 528. This proves (67); in particular the residual contains more than 4729 pivots throughout the interval.

The low-prime terms follow from the same two CRT progressions as in Lemma 6.24. Exhaustive modular inversion on the complete box $a, t \leq 5 \cdot 10^6$ gives, over $300 < p \leq 28540$, unrestricted prime-incidence maximum 383 and even non-base maximum 278. They occur at $t = 168$ and $t = 120$, respectively. For $p > 28540$, enumerate the complete quotient identity

$$p^2 r - 1 = at, \quad a, t \leq 5 \cdot 10^6.$$

There are 77829785 prime-quotient terms and 7742361 retained divisor events. Since $28540^4 > (5 \cdot 10^6)^2 + 1$, distinct witness primes cannot represent the same (a, t) ; the exact maximum degree is 49, at $t = 9240$. Prime 2 is inactive for even t , and prime 5 is inactive for every non-base pivot exactly as in the preceding lemma. The support calculation, middle-prime inversion, and complete tail enumeration are reproduced by `scripts/systematic_second_low_parity_close.py`, `scripts/systematic_low_high_prime_degree.cpp`, and `scripts/systematic_one_form_medium_exhaustion.cpp`. $\square$

Lemma 6.28 (Second-low parity diagonal ceilings). *On $3 \cdot 10^6 \leq N < 5 \cdot 10^6$, the nonsquarefree diagonal among the non-base classes is bounded by*

<i>allowed support</i>	<i>count</i>
<i>unrestricted</i>	$0.025295N$
<i>the two odd classes modulo 4</i>	$0.012655N$
<i>one odd class modulo 4</i>	$0.006333N$.

Proof. The complete eventwise Möbius-CRT producer on this interval again uses (51) and exact integer comparisons. Decimal renderings of its three largest rational envelopes are

$$0.0252947682147\dots, \quad 0.0126547423852\dots, \quad 0.0063325725290\dots$$

The asserted decimals are strict upward ceilings. The enumeration and the three assertions are in `scripts/systematic_concentrated_diagonal_blocks.cpp`. $\square$

Theorem 6.29 (Direct close of the second low block). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every*

$$3 \cdot 10^6 \leq N < 5 \cdot 10^6.$$

Proof. The pure opposite-base case is already closed by Theorem 6.11. In a mixed defect, Lemma 6.27 supplies more than 4729 non-base residual pivots, in particular more than the 22 needed below.

Apply the same exhaustive parity trichotomy as in Theorem 6.26. Eight even pivots give

$$|A| \leq 0.025295N + E_5(N).$$

If there are at most seven even pivots and both odd classes modulo 4 have at least eight members, choose four from each class; then

$$|A| \leq 0.012655N + 7 + 4O_5(N).$$

Otherwise charge the at most seven even pivots and the at most seven pivots in the smaller odd class. More than 22 residual pivots ensure that eight remain in the other odd class. Selecting them gives

$$|A| \leq 0.006333N + 14 + 2 \left\lceil \frac{N}{100} \right\rceil + 3O_5(N).$$

Thus no parity allocation is omitted.

Each right side is affine and increasing between its ceiling jumps. Checking the lower endpoint and every point immediately after a multiple of $25p^2$, $100p^2$, or 100 gives 29322 exact candidates and

branch	worst N	count margin
eight even	3000151	4901.940455
two odd classes	3000000	41259.720000
one odd class	3000801	10407.687267.

All three margins are against the unchanged Hall target $N/25 - 7/25$ and are strictly positive. The exact scan is `scripts/systematic_second_low_parity_close.py`, so every possible mixed defect is contradicted and the block is closed. $\square$

6.4 Kernel-checked high-tail certificate

We record the high endpoint in the same ten structural branches used in Proposition 6.18. Write these branches as

$$E_1G, E_{13}, E_2G, E_{23}, E_3G, E_{33}, O_2G, O_{23}, O_1G, O_{13}.$$

Lemma 6.30 (Global ten-branch allocation). *Let $N \geq 2 \cdot 10^7$ and let B be a compatible outside set with a strict Hall defect. Then at least one of the ten alternatives needed below may be selected: a generic or common-mod-9 triple in E_1, E_2 , or E_3 after charging the preceding sparse even classes, or a generic or common-mod-9 triple in two odd classes or one odd class after charging all three even classes.*

Proof. The cutoff-19 one-form certificate is uniform for $N \geq 2 \cdot 10^7$. For every actual residual pivot t it proves the exact integer inequality

$$4500000N + 10^9 |C_N^{18}| < 10^9 (|\{a \in C_N : at + 1 \text{ is squarefree}\}| + |\{a \in C_N^{18} : at + 1 \text{ is squarefree}\}|).$$

The proof uses the complete small-prime masks through 19, the reciprocal-square tail, and the transformed-root theorem; hence it applies to the literal neighbour sets, not to an abstract degree placeholder. The strict Hall defect and the opposite-base matching reduction then imply that the residual set has more than $.0045N$ members.

Put $g_N = N/20001 + 1$. Inspect E_1, E_2, E_3 in order. If the current class has more than g_N members, the gap lemma selects three pivots with the required span; they either are nonconstant modulo 9 or lie in one mod-9 cell, giving the two alternatives for that class. Otherwise charge the class, at cost at most g_N . If all three are charged, the odd remainder has more than

$$.0045N - 3(N/20001 + 1) > 3$$

members. Splitting it between the two odd classes, applying the same gap selection in the nonempty allocation, and then splitting by constancy modulo 9 gives the final four alternatives. These cases are disjoint at their first successful class and exhaustive. The literal degree reduction and allocation are kernel-checked in `lean4/Erdos848/TailTwentyMillionDegree19.lean` and `lean4/Erdos848/TailTwentyMillionBranchAllocation.lean`. $\square$

For a row with lower endpoint L , let

$$(d_U, d_C, d_{E_2}, d_L, d_{O_2}, d_{O_2C}, d_{O_1}, d_{O_1C})$$

be its eight certified diagonal envelopes, and let (r_1, r_2, r_3, r_O) be the four certified transformed-root envelopes. Put $\delta_L = 1/20001 + 1/L$. The residual payments in the displayed branch order are

$$d_U, d_C, d_U, d_{E_2} + \delta_L, d_L + 2\delta_L, d_L + 2\delta_L, d_{O_2} + 3\delta_L, d_{O_2C} + 3\delta_L, d_{O_1} + 3\delta_L, d_{O_1C} + 3\delta_L.$$

The finite small-prime payments are, in the same order,

$$10^{-6}(8685, 12616, 8685, 12616, 8685, 12616, 19420, 20878, 26643, 29459).$$

For the first six branches the remaining tail payment is

$$\frac{3}{25}\widehat{r}_{23} + \frac{r_i}{2}, \quad \widehat{r}_{23} = \frac{64081802747648035629863}{7596668444022826249000000},$$

with $r_i = r_1, r_2, r_3$ as appropriate. For the last four it is

$$\frac{124487826292585911291592581}{100465940172201877143025000000} + \frac{r_O}{2}.$$

Thus every row is accepted only after all ten exact rational totals are strictly less than

$$\frac{1}{25} - \frac{7}{25L}. \tag{68}$$

This is the complete arithmetic checker; it stores no theorem-valued field.

The finite transformed-root certificate uses split 125, modulus $M = 15625$, and the survivor vector

$$(6250, 6250, 6250, 6250, 6250, 3189, 1589, 789, 389, 190)$$

for support lengths $0, \dots, 9$. For each length it checks the product forcing and all four combinations of ordinary/five-twist and square/nonsquare mod-5 cosets. The QR word masks carry pointwise proofs of their residue semantics. Since the product of the first ten possible support primes is

$$742073813481 > 5 \cdot 10^{11},$$

every pivot in the finite high range has one of these ten support lengths. The checked survivor bound, the exact factors 1, 2, 4, 1 for E_1, E_2, E_3, O , and an anchored prime-count sieve give the four root envelopes. Independently, a grouped Möbius–CRT certificate gives all eight diagonal envelopes. The resulting six rows are:

$[L, U)$	prime profile	controlling branch	$\max_{\beta} T_{\beta}$	strict margin in (68)
$[2, 2.3)\text{B}$	16	E_{13}	$< .039752949176$	$> .000247050684$
$[2.3, 2.8)\text{B}$	18	E_{13}	$< .039579016176$	$> .000420983702$
$[2.8, 4.2)\text{B}$	18	E_2G	$< .039731687176$	$> .000268312724$
$[4.2, 6)\text{B}$	18	E_2G	$< .039896384176$	$> .000103615757$
$[6, 12)\text{B}$	30	E_{13}	$< .039780278676$	$> .000219721277$
$[12, 500)\text{B}$	30	E_{13}	$< .039710496176$	$> .000289503800$

The decimals are only outward renderings; the accepted comparisons are the exact rational inequalities above. Consecutive endpoints are checked as integers, so the six rows cover every $2 \cdot 10^9 \leq N < 5 \cdot 10^{11}$.

For $N \geq 5 \cdot 10^{11}$ the certificate is uniform. It uses split 120, root floor 840, the 47-prime anchored sieve, the common root envelope $8062340/10^9$, and diagonal envelopes

$$10^{-9}(26005710, 20944421, 17208292, 16649746, 13003280, 7437057, 6604680, 917036).$$

The corresponding grouped wheel divisors are

$$(1821, 2892, 2627, 2295, 3643, 4590, 3643, 3643).$$

The same exact checker has controlling branch E_1G , total $< .039734141676$, and strict margin $> .000265858323$.

Theorem 6.31 (Certified high-tail close). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every $N \geq 2 \cdot 10^9$.*

Proof. Assume a strict Hall defect and put $A = B \cup T_N(B)$. Lemma 6.30 places the defect in one of the ten branches above. If $2 \cdot 10^9 \leq N < 5 \cdot 10^{11}$, the endpoint cover selects one of the six finite rows. Its QR certificate bounds the transformed-root term, its grouped diagonal certificate bounds the selected diagonal term, and the branch checker bounds the normalized Hall completion by T_{β} . The strict row comparison and monotonicity of the target give

$$\frac{|B| + |T_N(B)|}{N} \leq T_{\beta} < \frac{1}{25} - \frac{7}{25L} \leq \frac{1}{25} - \frac{7}{25N},$$

contradicting the defect. For $N \geq 5 \cdot 10^{11}$ the identical argument uses the uniform row.

All implications just invoked are proved as ordinary propositions in

```
lean4/Erdos848/TailHighQrFiniteClose.lean
lean4/Erdos848/TailHighUnboundedClose.lean.
```

The six finite rows are in

```
lean4/Erdos848/GeneratedHybridPaperRootCoverage/
HighQrFinalRows.lean.
```

Their QR semantics are imported by that file. Ordinary Lean kernel replay of the assembled endpoint

reports only `propext`, `Classical.choice`, and `Quot.sound`; no external evaluator, theorem-valued certificate, or unchecked native decision is used. $\square$

Theorem 6.32 (Erdős Problem 848: terminal assembly). *For every integer $N \geq 1$, the maximum cardinality of a set $A \subseteq [1, N]$ for which $ab + 1$ is nonsquarefree for every $a, b \in A$ is*

$$|\{n \leq N : n \equiv 7 \pmod{25}\}|.$$

Equivalently, the Hall inequality (2) holds for every N .

Proof. Theorem 3.1 gives the upper bound for $N \leq 1.5 \cdot 10^6$. Theorems 6.26 and 6.29 give it on $[1.5 \cdot 10^6, 3 \cdot 10^6)$ and $[3 \cdot 10^6, 5 \cdot 10^6)$, respectively. Theorem 6.21 gives it directly for every $N \geq 5 \cdot 10^6$. These paper ranges cover all positive integers. Proposition 1.1 supplies the matching lower bound, and Theorem 2.2 gives the equivalent Hall formulation. $\square$

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